
GenExam: A Multidisciplinary Text-to-Image Exam

Zhaokai Wang^{1 2 *} Penghao Yin^{3 2 *} Xiangyu Zhao^{1 2} Changyao Tian^{4 2}
Yu Qiao² Wenhai Wang^{4 2} Jifeng Dai³ Gen Luo²

Abstract

Exams are a fundamental test of expert-level intelligence and require integrated understanding, reasoning, and generation. Existing exam-style benchmarks mainly focus on understanding and reasoning tasks, and current generation benchmarks emphasize the illustration of world knowledge and visual concepts, neglecting the evaluation of rigorous drawing exams. We introduce GenExam, the first benchmark for multidisciplinary text-to-image exams, featuring 1,000 samples across 10 subjects with exam-style prompts organized under a four-level taxonomy. Each problem is equipped with ground-truth images and fine-grained scoring points to enable a precise evaluation of semantic correctness and visual plausibility. Experiments on 17 text-to-image and unified models demonstrate the great challenge of GenExam and the huge gap where open-source models consistently lag behind the leading closed-source ones. By framing image generation as an exam, GenExam offers a rigorous assessment of models’ ability to integrate understanding, reasoning, and generation, providing insights for on the path to intelligent generative models. Our benchmark and evaluation code are released at <https://github.com/OpenGVLab/GenExam>.

1. Introduction

Exams are the ultimate test of expert-level intelligence. They are not merely about recalling knowledge points, but serve as a comprehensive assessment of understanding, reasoning, and generation. The ability to solve multidisciplinary exam problems indicates that a model has expert-level intelligence surpassing that of most adults. Therefore, exam-style bench-

marks naturally serve as a crucial yardstick to evaluate the progress of artificial general intelligence (AGI) towards domain expertise (Morris et al., 2024; Yue et al., 2024; Phan et al., 2025).

However, existing multidisciplinary benchmarks are primarily focused on understanding tasks, including pure text benchmarks (Wang et al., 2024b; Zhong et al., 2024) and multimodal benchmarks (Yue et al., 2024; 2025), but rarely focus on image generation tasks. In the field of text-to-image generation (T2I), current benchmarks focus more on general world knowledge reasoning (Niu et al., 2026; Sun et al., 2025). Some benchmarks have explored the domain of subject-specific images but are largely limited to concept illustration (Luo et al., 2025; Chang et al., 2025b;a), a relatively simple task with loose diagnostic criteria, similar to “illustrating concepts through image generation” rather than “solving a drawing exam”.

Upon a closer inspection of graph-drawing questions in common multidisciplinary exams, such as AP (CollegeBoard, 1952), A-level (Cambridge, 1951), and IB (IBO, 1968), we find that text-to-image exams present far distinct challenges against common image generation tasks:

1. The questions (prompts) are typically more complex, precise, and diverse, with strict and explicit constraints on drawing (generation);
2. Each question is equipped with a reference answer (ground truth image) and detailed scoring criteria (scoring points), enabling rigorous evaluation of the correctness of drawn images;
3. Knowledge scope is broader and deeper and can be systematically organized into a hierarchical structure (taxonomy);
4. Solving them demands rigorous semantic accuracy in drawn images, requiring integration of *understanding, reasoning, and generation*.

As shown in Fig. 1, even state-of-the-art text-to-image models, such as Nano Banana Pro (Google, 2025b) and GPT-Image-1.5 (OpenAI, 2025c) excel at producing images with superficially plausible overall structure but often fall short in portraying correct exam details.

Based on this motivation, we introduce *GenExam*, the first

*Equal contribution

¹Shanghai Jiao Tong University ²Shanghai AI Laboratory

³Tsinghua University ⁴The Chinese University of Hong Kong.
Correspondence to: Gen Luo <luogen@pjlab.org.cn>.

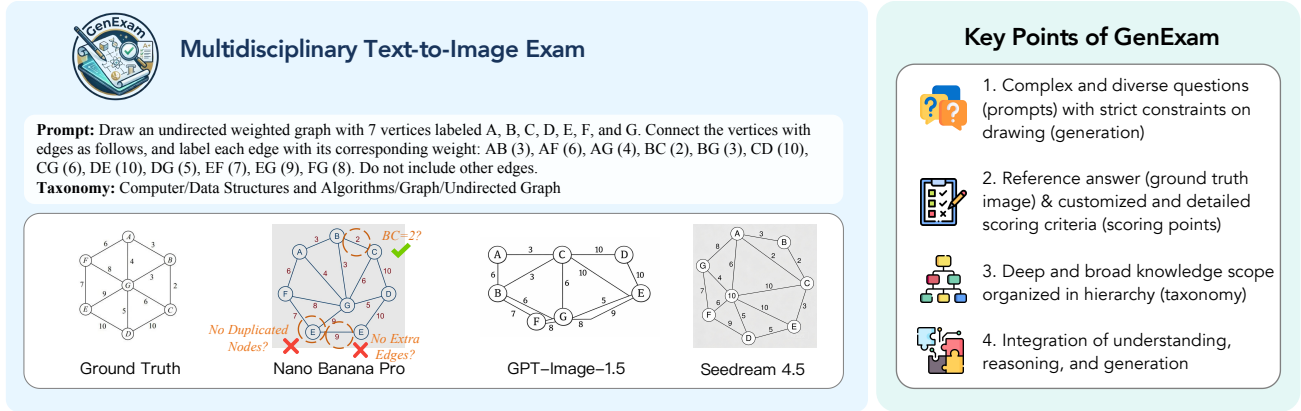


Figure 1. Examples of state-of-the-art generative models on the GenExam benchmark. Orange dashed circles indicate scoring points. GenExam contains complex and diverse prompts resembling human exams, and pose great challenge to existing models.

benchmark dedicated to multidisciplinary text-to-image exams. As shown in Fig. 2, GenExam asks the model to accomplish the subject-specific drawing exam with a series of detailed, rigorous and knowledgeable requirements. It includes 1,000 samples of 10 subjects with detailed four-level taxonomy, collected from college-level exams and open-source benchmarks and carefully reviewed by PhD annotators. In addition, GenExam adopts a strict prompt generation process through cooperation of GPT-5 (OpenAI, 2025a) and humans, with each prompt designed in the style of real exams (CollegeBoard, 1952; IBO, 1968). Given these designs, models are required to seamlessly integrate their understanding, reasoning and generation capabilities to solve the problem of GenExam.

The evaluation of multidisciplinary images also poses challenges, as they prioritize semantic correctness over photo-realism or aesthetic quality, different from natural images. Relying solely on a single instruction for MLLM-as-a-judge evaluation (Zhang et al., 2025c) makes it difficult to cover all requirements in the prompt. Therefore, we design customized evaluation criteria for each individual sample, similar to the exam marking process. As shown in in Fig. 1, scoring points of each sample are produced through a rigorous review process of GPT-5 and humans, with the ground truth image as the evaluation reference. Leveraging the most advanced commercial MLLM, *i.e.* GPT-5, we can automatically diagnose each scoring point through visual question answering. In addition to the correctness evaluation, GenExam also includes additional metrics to assess the visual plausibility, namely encompassing spelling, readability, and logical consistency. Based on the above metrics, we calculate two final scores to satisfy different evaluation requirements, with strict and relaxed standards respectively.

Our experimental results show that although Nano Banana Pro and GPT-Image-1.5 achieves strict scores of 70.2% and

42.5%, respectively, most models can only obtain strict scores less than 15%, and many latest T2I and unified multimodal large language models (MLLMs) like FLUX.2 dev (Labs, 2025), Qwen-Image-2512 (Wu et al., 2025) and BAGEL (Deng et al., 2025) even yield strict scores less than 3%. These results highlight the great challenge of GenExam and the huge gap between closed-source and open-source models, suggesting potential directions for existing models towards AGI. Overall, our contributions can be summarized in three folds:

- We propose GenExam, the first benchmark for multidisciplinary text-to-image exams, with carefully curated prompts and ground truth images. It aims to serve as a real exam to test whether models can seamlessly integrate understanding, reasoning and generation.
- We design an evaluation framework tailored for multidisciplinary images, assessing the semantic correctness and visual plausibility of disciplinary images through scoring points customized for each sample.
- Comprehensive experiments on existing T2I models and unified MLLMs highlight the challenge of GenExam. We provide analysis and insights on their capabilities and areas for improvement.

2. Related Work

2.1. Generative Models

Text-to-image generation (T2I) has been a primary research focus in generative AI in recent years. The dominant approaches are based on diffusion (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022; Labs, 2024) or autoregressive objects (Ramesh et al., 2021; Tian et al., 2024; Sun et al., 2024), which excel at generating images with high aesthetic quality and strong semantic consistency with

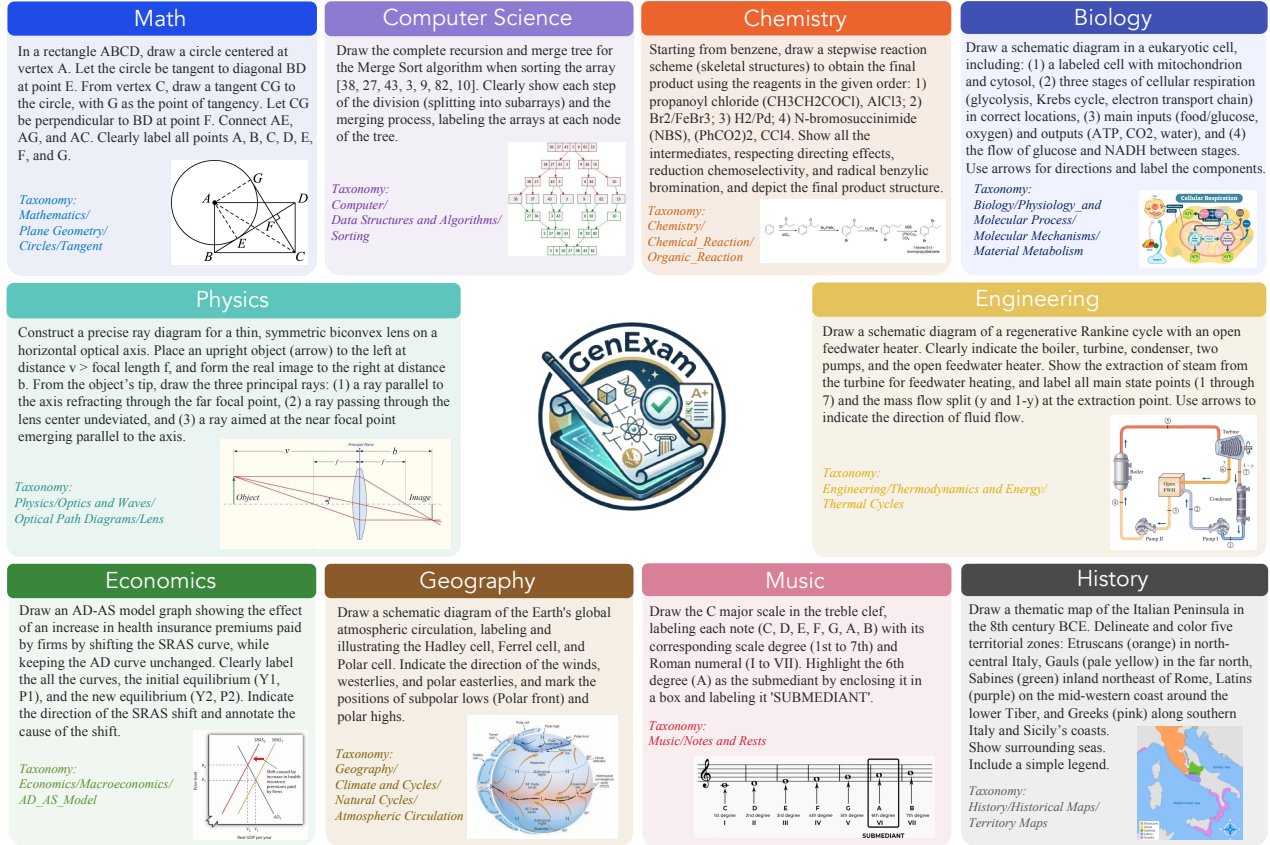


Figure 2. Examples from GenExam. GenExam contains 1,000 exam-style prompts that span over 10 subjects and corresponding reference images for multidisciplinary text-to-image exam.

prompts. Recently, unified MLLMs have emerged as a popular research topic, which supports multimodal understanding and image generation with a single model (Wang et al., 2024a; Deng et al., 2025; Xie et al., 2025; Li et al., 2025; Tian et al., 2026). The core advantage of unified MLLMs lies in providing powerful multimodal understanding and reasoning capabilities for image generation. The latest state-of-the-art models like GPT-Image-1.5 (OpenAI, 2025c) and Nano Banana Pro (Google, 2025b) show strong capabilities in generating plausible natural and aesthetic images. However, as shown in our experiments, their ability on multidisciplinary text-to-image exams is still limited.

2.2. Image Generation Benchmarks

Generation benchmarks (Zhao et al., 2025; Ghosh et al., 2023; Niu et al., 2026; Xia et al., 2025; Liu et al., 2026a; Li et al., 2026; Liu et al., 2026b) are of great importance in guiding models towards intelligent generative models. The early evaluation for image generation is mainly based on image similarity metrics (Heusel et al., 2017), which fails to capture the higher-level semantics. The current evaluation focuses primarily on literal prompt-image alignment,

with benchmarks like GenEval (Ghosh et al., 2023) and metrics like CLIP score (Hessel et al., 2021) and VQA score (Lin et al., 2024). Recent works move towards reasoning-informed generation (Niu et al., 2026; Zhao et al., 2025; Meng et al., 2024; Sun et al., 2025; Zhang et al., 2025a), and typically rely on MLLM-as-a-judge (Zhang et al., 2025c) for evaluation. However, their focus is mainly on world knowledge or commonsense reasoning, with domains restricted to natural or synthetic images.

Meanwhile, benchmarks such as MMMG (Luo et al., 2025), OneIG-Bench (Chang et al., 2025a), and SridBench (Chang et al., 2025b) explore multidisciplinary image generation, but primarily in the form of knowledge-concept illustration, a relatively simple task with loose diagnostic criteria compared to exam-style questions. Existing benchmarks for multidisciplinary exams are mainly for understanding tasks (Yue et al., 2024; Xia et al., 2025; Phan et al., 2025), but rarely focus on image generation. In contrast, our GenExam targets multidisciplinary text-to-image exams, with complex, precise, and diverse prompts and strict and explicit constraints on generation. By designing fine-grained scoring points, GenExam provides a more rigorous evaluation of reasoning-informed generation in multidisciplinary images.

3. GenExam

3.1. Overview

Fig. 2 provides an overview of our GenExam benchmark. It contains 1,000 samples that span over 10 subjects: mathematics, physics, chemistry, biology, computer science, geography, economics, music, history, and engineering. Each sample contains a ground truth image, a four-level taxonomy, and several scoring points.

Scoring Points. One critical challenge in the evaluation of exam-style image generation is that it is difficult to judge the semantic correctness of the generated image. This becomes evident when using MLLM-as-a-judge (Zhang et al., 2025c) with a single instruction template, where MLLMs often fail to cover all the requirements specified in the prompt. Examples include the specific structure of organic molecules (*e.g.*, the number of each type of atom, functional groups, chemical bonds), positional relationships in geometric figures, all notes in a musical score, etc. Therefore, inspired by scoring criteria for graph-drawing questions in human exams, we design several scoring points for each prompt, in the form of questions. By answering the questions, we can decide whether the generated image is correct, *e.g.* “Does the molecule contain exactly 8 carbon atoms?”. This ensures that each sample has a unique evaluation criterion, which improves the accuracy and stability of the evaluation. Each scoring point has a predefined score that sums up to 1, similar to the grading system in exams. The detailed evaluation protocol is given in Sec. 3.2.

Taxonomy. In addition to the top-level subject, we also provide a four-level taxonomy, such as “Mathematics/Plane Geometry/Circles/Tangent” in Fig. 2. The taxonomy annotations are built based on ISCED-F (UNESCO, 2013) fields, with the aim of improving the academic strictness of GenExam for seamless integration with multidisciplinary research. Detailed taxonomy is provided in Appendix D.

Statistics. In Tab. 1, the prompts in GenExam exhibit an average length of 74.8 words, providing dense and detailed requirements for the generated images, resembling difficult graph-drawing questions in real human exams. Each prompt has an average of 6.9 scoring points to thoroughly assess the generated image from all aspects based on the input prompt. In Fig. 3, we visualize the distributions of the image types, which demonstrate the diversity of images in our benchmark. We also provide a subset to facilitate an efficient evaluation. More statistics are provided in Appendix A.

Data Curation. We first consider possible subjects and domains of multidisciplinary images and curate a four-level taxonomy, and use it to generate keywords for web image collection. Images from existing subject-knowledge-related multimodal understanding datasets are also collected. We then conduct automatic filtering by utilizing GPT-5 to rate

images in terms of text richness, image domain, complexity and subject knowledge density, and set a threshold to remove undesirable images. Subsequently, prompts and scoring points are curated through GPT-5 drafting and rigorous human refinement. Detailed data curation pipeline is provided in Sec. B.

3.2. Evaluation Protocol

Unlike the evaluation of natural images, multidisciplinary images focus more on correctness than photorealism or aesthetic quality, and a simple flaw can cause significant errors. For instance, if a single atom is drawn mistakenly in a chemical structure or some arrows are depicted in the opposite direction in a diagram, the entire image is incorrect. In practice, common failure cases of current T2I models (Fig. 6) are either from semantic consistency with the prompt or plausibility of the image itself (*e.g.* spelling errors). Therefore, to thoroughly judge the correctness and overall quality of the generated image, we consider two dimensions: semantic correctness and visual plausibility. The evaluation pipeline is shown in Fig. 4.

Semantic Correctness. This dimension assesses the image’s consistency with the input prompt. The scoring points mentioned in Sec. 3.1 provide detailed criteria to judge each specific prompt. We use the MLLM judge to answer each scoring points questions by “Yes” or “No” like visual question answering. Since the answers should all be positive for a correct image, we can compute semantic correctness score as the sum of scores of all the questions whose answers are “Yes”. The range of semantic correctness is 0-1. Since judging certain subject knowledge-related aspects can be difficult, we add the ground truth image as input, so that the MLLM judge can use it as reference.

Visual Plausibility. This dimension focuses on the image itself, consisting of three sub-dimensions:

1. **Spelling:** Whether the spelling of the text in the image is correct, including notation and equations. This assesses the model’s text rendering capability, which has become the focus in recent models (Google, 2025b; Wu et al., 2025).
2. **Logical Consistency:** Whether all marks, text, musical notes, etc. are logically consistent, *e.g.* the labeled coordinates and the actual position. This evaluates the model’s ability to maintain internal coherence in the generated content, reflecting its capacity to integrate multiple components in a logically unified manner.
3. **Readability:** Whether all components are clearly readable and identifiable, without incorrectly placed labels and marks, overlap, occlusion, and unlabeled key components. This requires the model to design an appropriate layout and place the components, *e.g.* through reasoning.

Statistics	Value
Taxonomy Count (level 1/2/3/4)	10/40/132/236
Subject Knowledge Difficulty (easy:medium:hard)	24%: 38%: 38%
Minimum Number of Scoring Points	3
Maximum Number of Scoring Points	14
Average Number of Scoring Points	6.9
Minimum Prompt Length (words)	24
Maximum Prompt Length (words)	173
Average Prompt Length (words)	74.8

Table 1. Key statistics of GenExam.

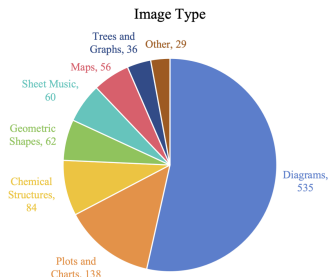


Figure 3. Distribution of image types.

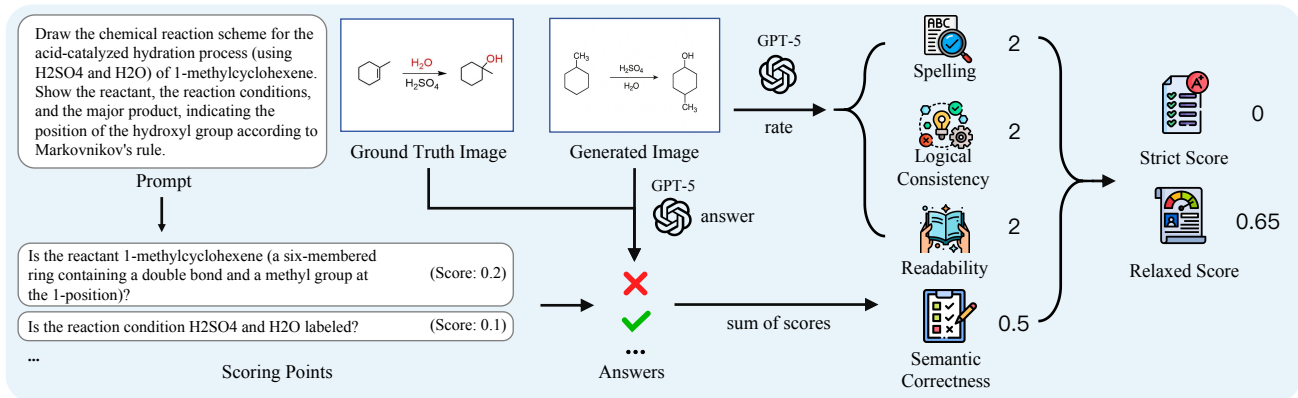


Figure 4. **Evaluation protocol of GenExam.** Employing MLLM-as-a-judge, we use scoring points of each prompt to calculate semantic correctness (0-1), and also rate the image in terms of spelling, logical consistency, and readability (0/1/2). The four dimensions are used to calculate a strict score and a relaxed score.

All the three sub-dimensions are in range 0-2, where 2 indicates almost perfect (tiny errors are allowed), 1 indicates a few flaws that slightly hinder the understanding of the key information, and 0 indicates critical errors that significantly hinder the understanding of the key information.

Strict and Relaxed Scores. Based on the two dimensions, we first calculate a strict score as the percentage of correct images among all generated images. An image is considered correct if and only if semantic correctness is 1 (*i.e.* all scoring points are correct) and the scores of spelling, logical consistency and readability are all 2, indicating full satisfaction of all evaluation dimensions, similar to RISEBench (Zhao et al., 2025). It is only more aligned with multidisciplinary image generation where a tiny flaw could lead to a technically wrong image, posing great challenges for T2I models.

However, as shown in Tab. 2, existing methods struggle to generate strictly correct images on GenExam. To highlight the difference between these models, we additionally calculate a relaxed score as the weighted average of the semantic correctness, spelling, logical consistency and readability. The weights are selected based on alignment with human preferences, similar to the process in WiScore (Niu

et al., 2026), and the values are 0.7, 0.1, 0.1 and 0.1 respectively. We encourage future research to report both strict and relaxed scores for comparison.

Evaluator Model. We utilize MLLM-as-a-judge (Zhang et al., 2025c) for evaluation with carefully crafted prompts, provided in Appendix E.3. Specifically, we adopt GPT-5 (OpenAI, 2025a), which shows strong multimodal understanding capabilities in multidisciplinary images. In Tab. 3 and Appendix Tab. 4, we also validate the robustness of Gemini-3-Flash and other MLLMs.

4. Experiments

In this section, we evaluate the performance of representative methods on our GenExam. We also validate the alignment between MLLM-as-a-judge and human evaluation, and offer analysis on the failure cases of existing methods. We use GPT-5 (OpenAI, 2025a) as the evaluator model and set the reasoning effort as “low”. More experiments are provided in Appendix C, including results categorized by subject knowledge difficulty and level-2 taxonomy, detailed scores on each dimension, and ablations on the evaluators.

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Model	Math		Phy		Chem		Bio		Geo		Comp		Eng		Econ		Music		Hist		Overall	
	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel
<i>Closed-source Models</i>																						
Nano Banana Pro	55.6	86.3	75.2	95.1	60.2	88.7	75.6	95.9	75.8	96.5	65.7	91.7	71.2	95.1	88.3	97.2	61.5	91.0	97.6	99.9	70.2	93.0
GPT-Image-1.5	26.5	65.8	46.0	85.4	39.0	78.1	56.4	91.9	60.6	92.5	36.3	75.8	44.1	86.4	42.9	85.5	29.2	70.8	51.2	90.9	42.5	81.5
GPT-Image-1	7.9	51.9	13.3	66.4	13.6	53.4	22.8	73.7	15.9	73.9	10.3	55.6	13.1	65.5	13.0	65.8	9.2	52.7	2.4	67.4	13.1	62.2
Seedream 4.5	5.3	44.7	11.5	63.4	7.6	48.9	25.0	75.8	12.1	67.6	12.7	57.9	15.3	69.7	15.6	67.3	3.1	38.0	4.9	55.0	12.3	59.5
FLUX.2 max	6.6	49.0	8.8	63.2	6.8	54.0	11.6	74.5	15.2	76.3	8.8	56.5	10.8	68.9	2.6	61.5	6.2	47.0	7.3	68.0	8.6	61.6
Seedream 4.0	2.6	39.8	3.5	49.0	5.9	46.1	18.6	71.0	10.6	65.1	6.9	52.2	11.7	60.0	5.2	56.0	0.0	34.5	7.3	56.7	7.8	53.2
Imagen-4-Ultra	2.6	35.9	9.7	57.4	9.3	44.5	14.7	68.1	7.6	66.9	2.9	40.1	12.6	65.6	9.1	59.7	0.0	38.4	0.0	57.8	7.8	53.0
<i>Open-source T2I Models</i>																						
FLUX.2 dev	2.6	31.6	1.8	42.7	4.2	33.2	3.8	54.8	3.0	62.6	1.0	31.1	2.7	48.9	1.3	43.6	0.0	33.4	0.0	47.5	2.4	42.3
Qwen-Image-2512	0.0	27.9	2.7	41.3	0.8	23.2	1.3	44.4	6.1	56.6	0.0	24.1	4.5	42.9	0.0	32.3	0.0	28.3	0.0	37.0	1.5	35.3
HiDream-I1-Full	0.0	16.7	0.0	17.7	0.0	13.5	0.0	27.3	0.0	36.2	0.0	15.4	0.0	24.4	0.0	18.8	0.0	21.3	0.0	31.8	0.0	21.2
Stable Diffusion 3.5 Large	0.0	12.2	0.0	13.2	0.0	10.7	0.0	21.8	0.0	38.8	0.0	6.6	0.0	16.3	0.0	8.0	0.0	24.1	0.0	18.0	0.0	15.9
<i>Open-source Unified MLLMs</i>																						
BAGEL (thinking)	0.0	11.7	0.0	13.8	0.0	11.9	0.0	15.2	0.0	28.5	0.0	6.2	0.0	10.7	0.0	6.3	0.0	14.7	0.0	16.0	0.0	12.9
BAGEL	0.0	14.7	0.0	10.6	0.0	7.9	0.0	10.8	0.0	24.5	0.0	6.8	0.0	10.2	0.0	5.3	0.0	13.7	0.0	14.4	0.0	11.4
Show-o2-7B	0.0	10.8	0.0	11.9	0.0	4.8	0.0	12.8	0.0	33.3	0.0	4.7	0.0	11.8	0.0	7.0	0.0	8.8	0.0	14.5	0.0	11.2
Show-o2-1.5B-HQ	0.0	7.3	0.0	7.5	0.0	6.2	0.0	15.0	0.0	25.3	0.0	4.3	0.0	9.3	0.0	7.3	0.0	7.6	0.0	19.8	0.0	10.1
BLIP3o-NEXT-GRPO-Text-3B	0.0	15.5	0.0	10.5	0.0	9.2	0.0	15.5	0.0	23.7	0.0	8.2	0.0	10.1	0.0	8.1	0.0	15.2	0.0	10.2	0.0	12.6
BLIP3o-8B	0.0	6.4	0.0	5.5	0.0	4.7	0.0	7.0	0.0	16.7	0.0	3.6	0.0	8.4	0.0	2.5	0.0	6.0	0.0	11.2	0.0	6.7
Janus-Pro	0.0	13.7	0.0	8.8	0.0	8.2	0.0	7.2	0.0	18.8	0.0	3.9	0.0	10.5	0.0	4.2	0.0	14.5	0.0	6.6	0.0	9.5
Emu3	0.0	11.3	0.0	0.6	0.0	0.6	0.0	5.6	0.0	34.6	0.0	5.1	0.0	16.5	0.0	1.9	0.0	5.8	0.0	6.2	0.0	8.8

Table 2. Strict scores (Str) and relaxed scores (Rel) on GenExam.

4.1. Baselines

We select 17 representative models for comparison, including closed-source proprietary models, open-source T2I models and unified MLLMs. We use the default T2I configuration for each of the models for inference.

Closed-source Models: Nano Banana Pro (Google, 2025b), GPT-Image-1.5 (OpenAI, 2025c), GPT-Image-1 (OpenAI, 2025b), Seedream 4.5 (Seed, 2025), FLUX.2 max (Labs, 2025), and Imagen-4-Ultra (Deepmind, 2025). These proprietary models represent the state-of-the-art methods in text-to-image generation.

Open-source T2I Models: Qwen-Image-2512 (Wu et al., 2025), FLUX.2 dev (Labs, 2025), HiDream-I1-Full (Cai et al., 2025), and Stable Diffusion 3.5 Large (Rombach et al., 2022). These models are dedicated to the T2I task.

Open-source Unified MLLMs: Show-o2-7B, Show-o2-1.5B-HQ (Xie et al., 2025), BAGEL (Deng et al., 2025) (thinking & non-thinking), Janus-Pro (Chen et al., 2025c), Emu3 (Wang et al., 2024a), BLIP3o-8B (Chen et al., 2025a), and BLIP3o-NEXT-GRPO-Text-3B (Chen et al., 2025b). These models employ a unified architecture for T2I and multimodal understanding.

4.2. Main Results

Results on GenExam are provided in Tab. 2. The strict scores demonstrate the challenge of our benchmark. While Nano Banana Pro and GPT-Image-1.5 show relatively strong performance of 70.2% and 42.5%, other closed-source methods like Seedream 4.5 and FLUX.2 max can only achieve strict scores less than 15%. For open-source models, even the latest Qwen-Image-2512 and FLUX.2 dev fail in almost all cases, with strict scores less than 5%. This shows a huge gap between open-source and closed-source models in multidisciplinary text-to-image exams, suggesting potential direction for future improvement.

The differences between the models are highlighted with relaxed scores. The performance gap between open-source and closed-source models still exists. FLUX.2 dev and Qwen-Image-2512 achieve the highest relaxed scores of 42.3 and 35.3 among open-source methods. However, current unified MLLMs like BAGEL and Show-o2 still lag behind dedicated T2I models, suggesting a significant challenge for unified MLLMs to utilize its multidisciplinary knowledge and reasoning ability into image generation.

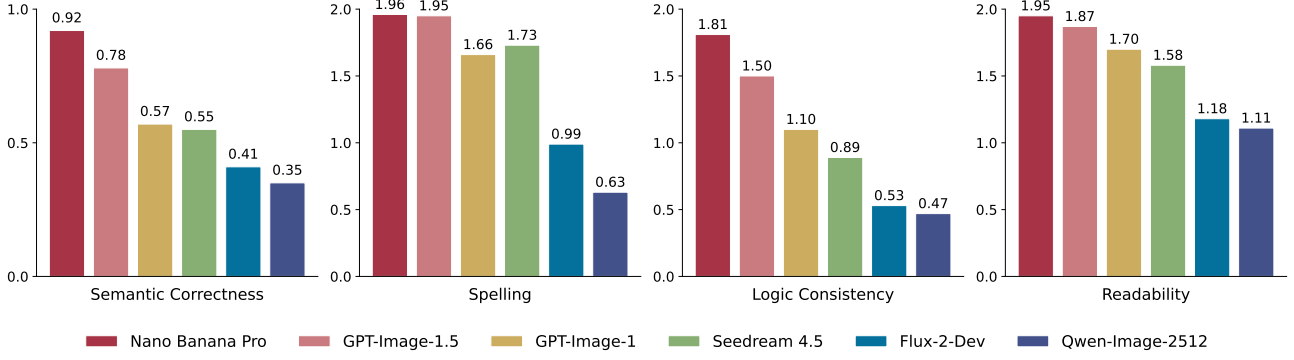


Figure 5. Comparison across models on four evaluation dimensions.

Metric	Kendall		Spearman		Pearson	
	τ	p	ρ	p	r	p
Relaxed Score By GPT-5	0.6746	<0.0001	0.8217	<0.0001	0.8444	<0.0001
Relaxed Score By Gemini-3-Flash	0.6608	<0.0001	0.8200	<0.0001	0.8261	<0.0001
Semantic Correctness	0.6333	<0.0001	0.7862	<0.0001	0.8062	<0.0001
VQA Score	0.1452	0.0009	0.2189	0.0005	0.1786	0.0046
CLIP Score	0.1158	0.0072	0.1620	0.0103	0.1647	0.0091

(a) MAE				
	Semantic Correctness	Spelling	Logical Consistency	Readability
Range	0-1	0-2	0-2	0-2
MAE	0.10	0.11	0.28	0.20

(b) Correlation						
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Table 3. Mean absolute error (MAE) and correlations between human preferences and automatic evaluation.

4.3. Human Alignment

We conduct a human alignment experiment to examine the validity of MLLM-as-a-judge, following the paradigms in (Zhao et al., 2025; Sun et al., 2025). Five experts are asked to score 250 randomly sampled images generated by GPT-Image-1.

They first provide an overall rating between 1 and 10, and then follow the automatic evaluation protocol to annotate scoring points, spelling, logical consistency and readability. The average scores among all experts are used to calculate the mean absolute error (MAE) and correlations between MLLM-judge predicted scores and human scores. In Tab. 3(a), the results show a low MAE across the four dimensions, indicating high precision of our evaluation.

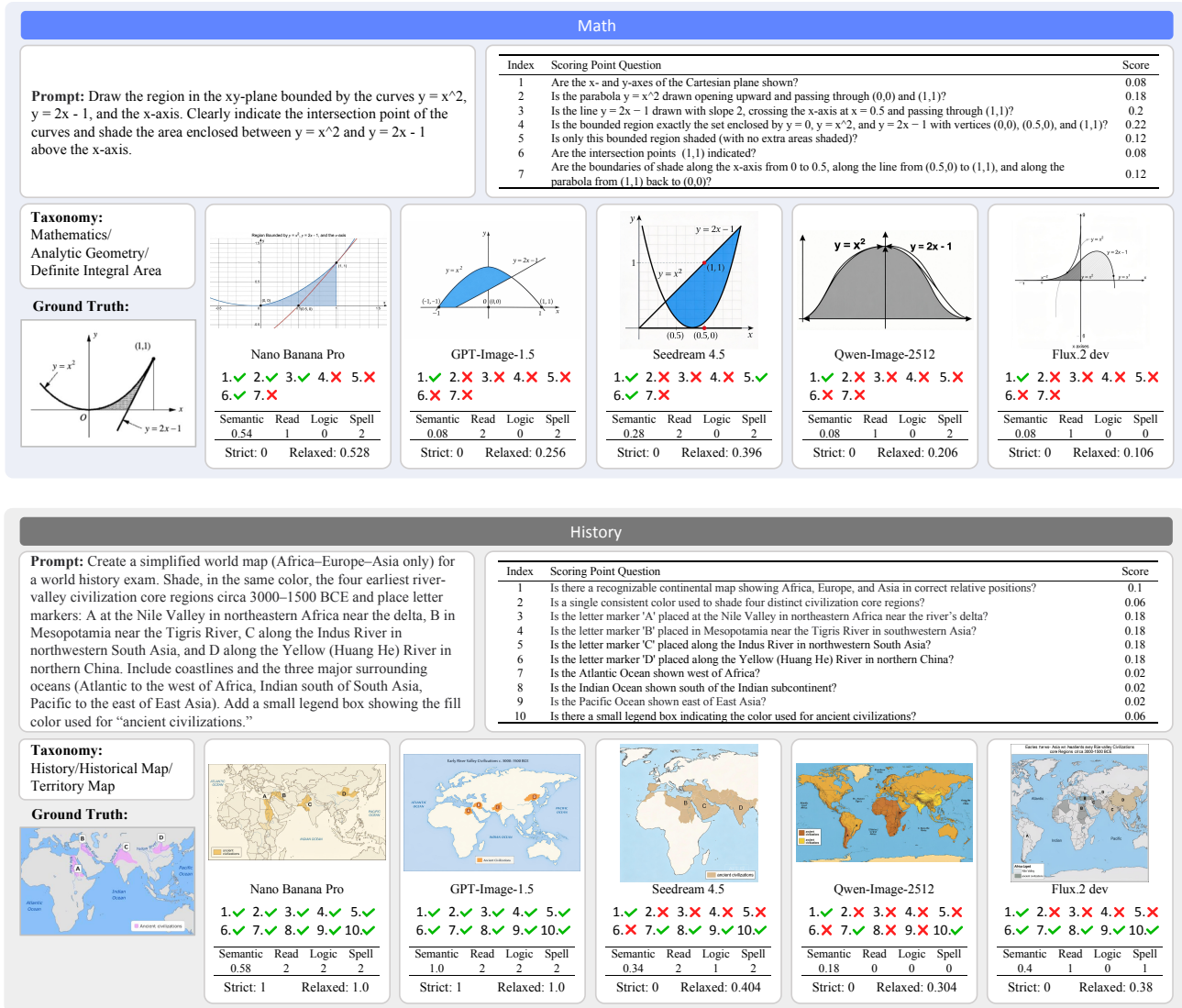
We then use the overall rating to calculate correlations (Kendall’s τ , Spearman’s ρ and Pearson’s r) between human scores and four automatic metrics: our relaxed score, our semantic correctness, VQA score (Lin et al., 2024), and CLIP score (Hessel et al., 2021). In Tab. 3(b), our relaxed score evaluated by GPT-5 achieves high correlations with human judgments with $p < 0.05$, slightly better than using solely semantic correctness without visual plausibility, while CLIP score and VQA score fail to capture the correctness of multidisciplinary images. When using Gemini-3-Flash (Google, 2025a) as the evaluator for relaxed scores, the results still show strong human alignment.

In Appendix Tab. 4, we further examine other closed-source and open-source evaluators. In Appendix Tab. 5, we extend to generated images of five other models to confirm strong correlations with human ratings across all models, and provide standard deviations to demonstrate the stability of the evaluation.

4.4. Error Analysis

To gain deeper insights into the strengths and limitations of the models, we provide a comparison of representative models on each evaluation dimension in Fig. 5. Specifically, Nano Banana Pro and GPT-Image-1.5 surpass other models across all four dimensions, notably with near-perfect spelling and readability, and Nano Banana Pro shows even stronger semantic correctness and logic consistency than GPT-Image-1.5. Open-source models Qwen-Image-2512 and FLUX.2 dev lag behind closed-source models across all dimensions, especially on spelling and logic consistency, suggesting that open-source models should first focus on basic drawing before tackling high-level semantics and reasoning.

Examples of generated images are shown in Fig. 6. As shown, state-of-the-art closed-source models such as Nano Banana Pro can generate relatively high-quality images that adheres to prompt specifications, despite some minor flaws. In contrast, open-source models often struggle with fundamental rendering skills, frequently generating incoherent


History

Prompt: Create a simplified world map (Africa–Europe–Asia only) for a world history exam. Shade, in the same color, the four earliest river-valley civilization core regions circa 3000–1500 BCE and place letter markers: A at the Nile Valley in northeastern Africa near the delta, B in Mesopotamia near the Tigris River, C along the Indus River in northwestern South Asia, and D along the Yellow (Huang He) River in northern China. Include coastlines and the three major surrounding oceans (Atlantic to the west of Africa, Indian south of South Asia, Pacific to the east of East Asia). Add a small legend box showing the fill color used for “ancient civilizations.”

Taxonomy:
History/Historical Map/
Territory Map

Ground Truth:



Nano Banana Pro

1.✓ 2.✓ 3.✓ 4.✓ 5.✓
6.✓ 7.✓ 8.✓ 9.✓ 10.✓

Semantic	Read	Logic	Spell
0.58	2	2	2

Strict: 1 Relaxed: 1.0

GPT-Image-1.5

1.✓ 2.✓ 3.✓ 4.✓ 5.✓
6.✓ 7.✓ 8.✓ 9.✓ 10.✓

Semantic	Read	Logic	Spell
1.0	2	2	2

Strict: 1 Relaxed: 1.0

Seedream 4.5

1.✓ 2.✗ 3.✗ 4.✗ 5.✗
6.✗ 7.✓ 8.✓ 9.✓ 10.✓

Semantic	Read	Logic	Spell
0.34	2	1	2

Strict: 0 Relaxed: 0.404

Qwen-Image-2512

1.✓ 2.✗ 3.✗ 4.✗ 5.✗
6.✗ 7.✓ 8.✗ 9.✗ 10.✓

Semantic	Read	Logic	Spell
0.18	0	0	0

Strict: 0 Relaxed: 0.304

Flux.2 dev

1.✓ 2.✗ 3.✗ 4.✗ 5.✗
6.✓ 7.✓ 8.✓ 9.✓ 10.✓

Semantic	Read	Logic	Spell
0.4	1	0	1

Strict: 0 Relaxed: 0.38

Figure 6. Examples of images generated by top-performing models. Semantic, Read, Logic, and Spell denote semantic correctness, logical consistency and spelling, respectively.

figures characterized by legibility errors (e.g., the blurred label “C” in Qwen-Image-2512 in History), logical inconsistencies (e.g., redundant axes in Qwen-Image-2512 Math), and spelling inaccuracies (e.g., “Ea?ies ?e’ws” in FLUX.2 dev in History). These visual defects not only diminish visual plausibility scores but also impede semantic evaluation.

It is worth noting that the inferior performance of open-source models often results not from a lack of necessary disciplinary knowledge, but from an inability to accurately translate that knowledge into readable visual figures. For instance, in the result of FLUX.2 dev in history, the model correctly identifies the regions of Egypt, Iran, India, and China (although with excessive highlighting) but fails to place the appropriate graphics within those regions. This

discrepancy suggests that the bottleneck lies not in the lack of knowledge, but in the failure to execute visual reasoning based on that knowledge. Consequently, enhancing the ability to transform internal knowledge into visual outputs and improving the alignment between conceptual understanding and generative execution could improve open-source models to a new stage of performance. These specific failures further underscore the value of GenExam’s fine-grained scoring criteria, which effectively capture nuanced flaws that broad metrics such as photorealism or general aesthetics would typically overlook. Additional qualitative results are provided in Appendix F.

5. Conclusion

We presented GenExam, the first benchmark for multidisciplinary text-to-image exams, designed to test whether models can integrate understanding, reasoning and generation to truly solve graph-drawing problems. GenExam offers diverse and challenging prompts with ground truth images and fine-grained scoring points, reflecting the rigor of real exams. Our comprehensive experiments on 17 existing models demonstrate the difficulty of the benchmark and the gap between closed-source and open-source models. We hope GenExam will guide the advancement of generative models toward expert-level intelligence in a multidisciplinary direction.

Impact Statement

This paper presents GenExam, a benchmark for evaluating multidisciplinary text-to-image generation models. Our goal is to advance research on generative models that integrate understanding, reasoning, and generation capabilities. Potential societal impacts include improving the evaluation and reliability of generative AI systems in educational and scientific domains. At the same time, increasingly capable generative models may also raise concerns regarding misuse, misinformation, or overreliance on AI-generated content. We hope GenExam can support future research on more accurate, reliable, and trustworthy generative models.

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A. More Statistics

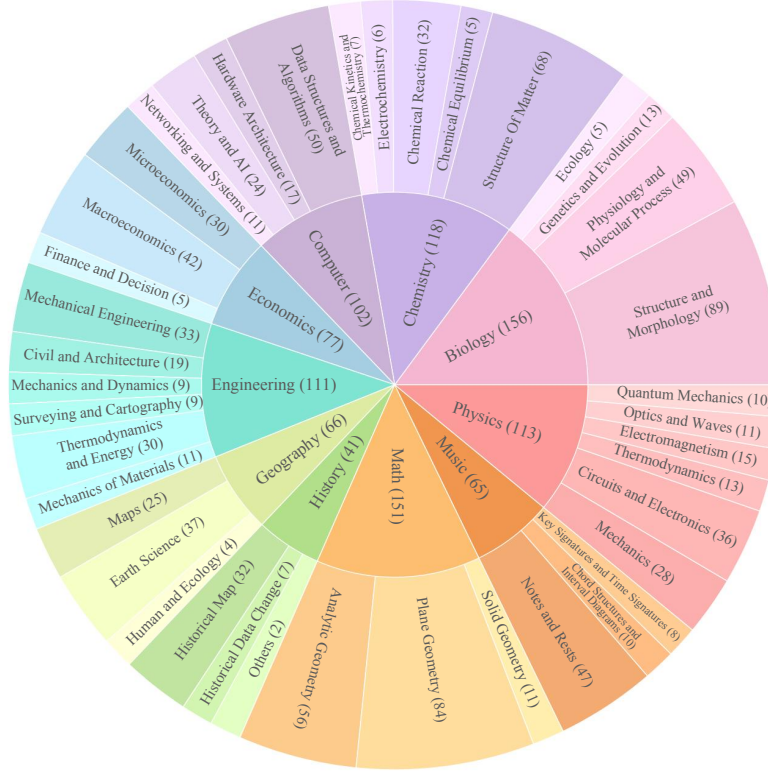


Figure 7. **Level-2 taxonomy of GenExam.** Detailed four-level taxonomy is in Sec D.

Fig. 7 shows all the level-2 taxonomy in GenExam, and Fig. 8(a) and (b) show the distribution of subjects in GenExam-Full and GenExam-Mini. We include more samples for science and engineering subjects while also covering subjects related to humanity and social science, such as Economics, Music, and History. GenExam-Mini is constructed using stratified sampling on level-3 taxonomy, which ensures similar distributions between GenExam-Full and GenExam-Mini.

The proportion of image sources is presented in Fig. 8(c), demonstrating the diversity of images in our benchmark. Word clouds of prompts in each subject are shown in Fig. 9, where we observe keywords with dense subject knowledge similar to human exams.

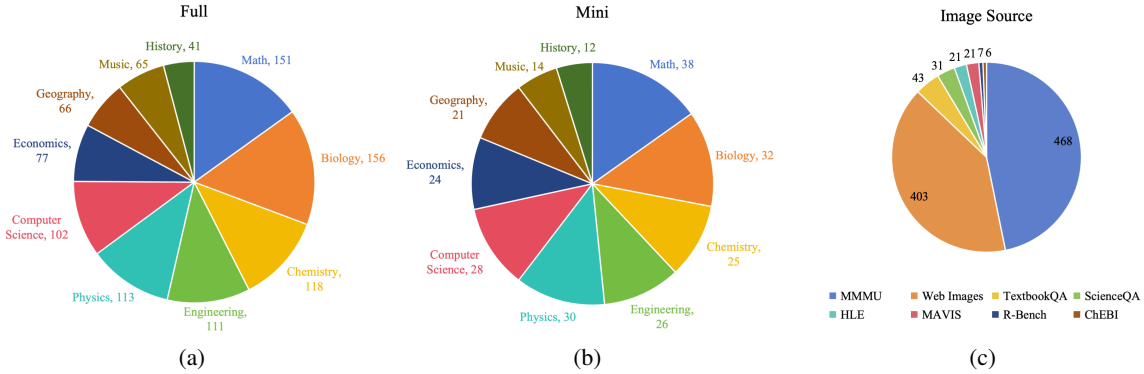
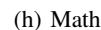
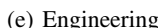


Figure 8. **More statistics:** (a) Subject distribution on GenExam-Full; (b) Subject distribution on GenExam-Mini; (c) Proportion of each image source.



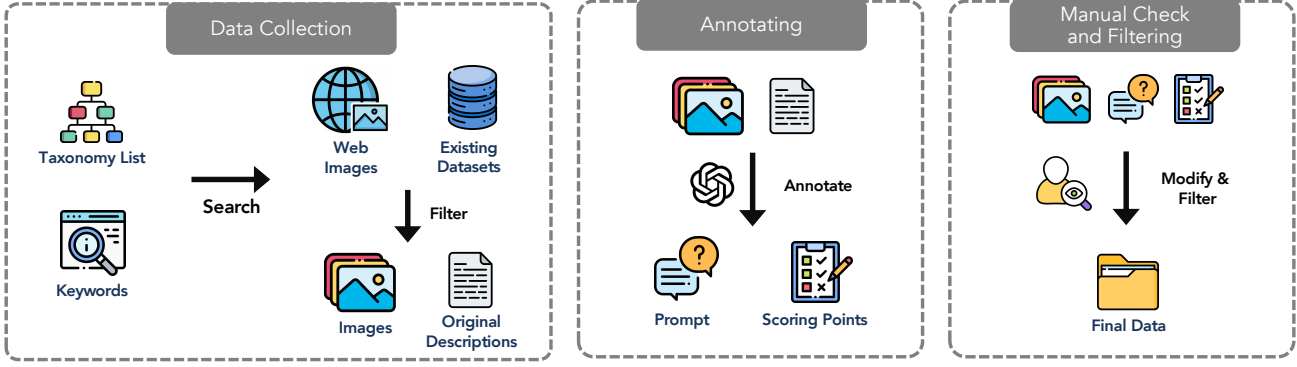


Figure 10. **Data curation pipeline.** We use pre-defined taxonomy to collect web images and existing datasets, and conduct annotating and filtering based on GPT-5 and manual check.

B. Data Curation Pipeline

Data Collection. The data curation pipeline is shown in Fig. 10. To construct a general benchmark for multidisciplinary text-to-image exam, we first consider possible subjects and domains of multidisciplinary images and curate a four-level taxonomy assisted by GPT-5 (OpenAI, 2025a) and through manual check. We then use these taxonomy to generate keywords related to specific subject knowledge, and collect web images from exams and textbooks based on Google search. Images from existing subject-knowledge-related datasets are also collected: MMMU (Yue et al., 2024), ScienceQA (Lu et al., 2022), TextbookQA (Kembhavi et al., 2017), MAVIS (Zhang et al., 2025b), ChEBI (Hastings et al., 2016), HLE (Phan et al., 2025) and R-Bench (Guo et al., 2025). We obtain 40K images from the data sources with their original text descriptions (e.g. webpage content for web images, lecture notes in TextbookQA, or question and answers in MMMU).

Automatic Filtering. To obtain a high-quality dataset, we employ an automatic filtering process on the collected images. We first design heuristic rules to remove images with duplicated content, low resolution, watermarks or non-English text. Then, we leverage GPT-5 to rate each image in terms of its text richness, image domain, image complexity, and subject knowledge density. We set a threshold for each of these aspects to filter out images that do not meet the requirements of our task, including those with:

- High text richness: e.g. tables or screenshots of pure text. Such images are dominated by textual information and primarily examines the text rendering ability, which deviates from the core requirement of more general “image generation” for visual content.
- Undesirable image domain: e.g. photographs, pathological images or satellite images. These domains rely more on realistic details or professional equipment for collection, which is inconsistent with real graph-drawing questions in exams. They hardly reflect the ability to visually transform interdisciplinary knowledge.
- High complexity: e.g. too many components or sub-figures. Excessive components or sub-figures lead to redundant visual information, which interferes with the model’s generation of core visual elements. Additionally, complex structures often exceed the model’s current generation capabilities and pose overly high challenges to the model.
- Low subject knowledge density: e.g. simple plots and charts without subject knowledge. Such images lack core information in interdisciplinary scenarios.

We obtain 6.5K images from automatic filtering for annotating.

Annotating. We employ GPT-5 to generate the initial version of prompts and scoring points. Prompts that require subject knowledge and reasoning are generated, similar to graph-drawing questions in exams. The original text description from the data source is used for reference to improve the precision, but should be treated carefully since they may contain irrelevant information. Subsequently, scoring points (questions and scores) are generated based on the prompt. We enforce through prompt engineering that the questions cannot be too general (e.g. “Does the image show the correct structure of benzene?”) or too complex (can be split into multiple sub-questions). Scores are based on the difficulty of the questions, where basic questions should have low scores and harder ones have high scores. We provide some few-shot examples to enhance the labeling process. The complete prompts are provided in Appendix E.2.

Manual Check and Filtering. Three PhD annotators perform rigorous manual inspection on all the images, prompt, and

Evaluator Model	Ground Truth	Scoring Points	Plausibility	Kendall	Spearman	Pearson	API Cost	Time
	Input	Accuracy $\uparrow$	MAE $\downarrow$	$\tau \uparrow$	$\rho \uparrow$	$r \uparrow$	(\$/img)	(s/img)
<i>Closed-source MLLMs</i>								
Gemini-3-Flash	Img	86.4	-	0.6608	0.8300	0.8261	0.063	27
GPT-5 + low reasoning effort	Img	88.5	0.20	0.6746	0.8217	0.8444	0.0242	72
GPT-5 + low reasoning effort	—	86.2	0.23	0.6141	0.7753	0.7976	0.0231	59
GPT-5 + low reasoning effort	Text	87.8	0.21	0.6559	0.8103	0.8336	0.0234	63
GPT-5 + low reasoning effort	Img, Text	88.5	0.20	0.6693	0.8345	0.8427	0.0250	75
GPT-5 + minimal reasoning effort	Img	84.2	0.28	0.6066	0.7521	0.7860	0.0144	35
GPT-5 + medium reasoning effort	Img	86.7	0.21	0.6617	0.8224	0.8358	0.0389	101
GPT-5 + high reasoning effort	Img	87.6	0.20	0.6647	0.8190	0.8304	0.0654	186
GPT-5-mini + minimal reasoning effort	Img	79.0	0.35	0.5446	0.6996	0.7185	0.0039	11
GPT-5-mini + low reasoning effort	Img	84.3	0.29	0.6002	0.7729	0.7806	0.0052	19
GPT-5-mini + medium reasoning effort	Img	84.2	0.25	0.6313	0.8025	0.8165	0.0079	36
GPT-5-mini + high reasoning effort	Img	83.4	0.26	0.5906	0.7611	0.7811	0.0196	119
o3	Img	82.0	0.23	0.6082	0.7706	0.7895	0.0229	39
o4-mini	Img	84.7	0.35	0.6300	0.7915	0.8092	0.0156	31
GPT-4.1	Img	71.2	0.39	0.4417	0.5745	0.5723	0.0142	29
GPT-4o	Img	68.5	0.39	0.4303	0.5700	0.5610	0.0134	23
<i>Open-source MLLMs</i>								
Qwen3-VL-235B-A22B-Thinking	Img	75.1	0.38	0.4606	0.6021	0.5759	-	105
Qwen3-VL-235B-A22B-Instruct	Img	71.6	0.43	0.4279	0.5703	0.6182	-	35
InternVL3.5-241B-A28B (thinking)	Img	63.3	0.41	0.2856	0.3538	0.3866	—	29
Intern-S1 (241B, thinking)	Img	57.0	0.51	0.0664*	0.0833*	0.2088*	—	39
Intern-S1-mini (8B, thinking)	Img	56.6	0.53	0.0027*	0.0023*	0.0158*	—	18

Table 4. **Ablation on the evaluator model.** Settings are identical to Tab. 3. *: The p -value is larger than 0.05, indicating the metric using this evaluator has no statistical significance.

scoring points, and cross-validate each other’s decision. Images with duplicate domain or undesirable text richness and complexity not recognized by automatic filtering are first manually removed. Subsequently, imprecise prompts and are manually modified to align with the ground truth images and include sufficient disciplinary knowledge. Inaccurate scoring points are removed or updated to match the prompt. We finally obtain 1K images as the final dataset.

C. Additional Experiments

C.1. Ablation on Evaluator Models

We further study the selection of evaluator model by examining different MLLMs, including closed-source models (Gemini-3-Flash (Google, 2025a), GPT-5, GPT-5-mini, o3, o4-mini, GPT-4.1, GPT-4o (OpenAI, 2025a;d)) and open-source models (Qwen3-VL (Bai et al., 2025b), InternVL3.5 (Wang et al., 2025), Intern-S1, Intern-S1-mini (Bai et al., 2025a)). On the same manually evaluated samples as in Tab. 3, we run each of the models as evaluator and calculate the accuracy of answering scoring points, MAE of visual plausibility, and three correlation metrics (Kendall’s τ , Spearman’s ρ , and Pearson’s r).

As shown in Tab. 4, the latest model GPT-5 with low reasoning effort achieves the best performance with acceptable cost and time, while using medium or high reasoning effort leads to a potential decrease in performance and higher costs. In addition, using ground truth images as reference help to improve the evaluation precision. Early closed-source models have relatively low performance, especially those without reasoning abilities.

For open-source models, Qwen3-VL demonstrates relative strong performance, but still lags behind GPT-5. Other models have a larger gap between closed-source models, where some scoring point accuracy is close to random guess (50%) and some correlations are even of no statistical significance, suggesting these evaluators are not applicable. This is because multidisciplinary images are still difficult for current multimodal understanding models (Yue et al., 2025). Therefore, it is necessary to use advanced MLLMs to evaluate the generated images for higher precision and robustness.

Model	Semantic		Spelling		Logic		Readability		Kendall τ	Spearman ρ	Pearson r
	MAE	std	MAE	std	MAE	std	MAE	std			
Nano Banana Pro	0.068	0.0142	0.071	0.0448	0.158	0.0573	0.118	0.0451	0.7246	0.8621	0.8893
GPT-1.5-Image	0.082	0.0169	0.086	0.0537	0.191	0.0692	0.143	0.0548	0.7013	0.8416	0.8679
Seedream 4.5	0.108	0.0224	0.113	0.0701	0.261	0.0946	0.195	0.0759	0.6672	0.8129	0.8248
FLUX.2 dev	0.126	0.0258	0.132	0.0817	0.301	0.1085	0.225	0.0853	0.6461	0.7928	0.7986
Qwen-Image-2512	0.134	0.0273	0.140	0.0864	0.317	0.1149	0.238	0.0901	0.6319	0.7784	0.7816

Table 5. Human alignment on images generated by more models.

We observe that using ground truth images as reference helps to improve the evaluation precision. We also test using captions of the ground truth images generated by GPT-5 as the ground truth, and observe that text-only ground truth is slightly worse than image-only ground truth. This is because ground truth images have better spatial/detail fidelity than textual descriptions (which often omit critical spatial details) and lower ambiguity. Using both images and textual descriptions shows no significant improvement than image-only, as images already contain sufficient detailed information and incorporating text becomes unnecessary.

C.2. Detailed Comparison of Existing Methods

In Tab. 5, we extend the human alignment experiment to 500 additional samples covering five other models. The results show that relaxed scores maintain strong correlations with human ratings across all models, confirming the metric’s generalizability. The standard deviation (std) values demonstrate high inter-rater reliability.

In Tab. 6 (a), we show the results on GenExam-Mini. The performance of the models is consistent with the results from the full dataset, which can be considered as a valid subset for efficient and affordable validation.

Results categorized by subject knowledge difficulty are presented in Tab. 6 (b). We observe that models tend to have lower performance on samples with higher subject knowledge difficulty, demonstrating the challenge of integrate profound multidisciplinary knowledge into generation.

We provide results categorized by image types in Tab. 7. We observe that models tend to have lower performance on samples with certain image types like chemical structures, geometric shapes, sheet music and trees & graphs. These suggest potential directions for future improvements.

The results on each level-2 taxonomy are provided in Tab. 8. To highlight the difference between models, and since some models can only obtain a 0% strict score, we only report relaxed scores.

Tab. 9 shows the scores of each dimension (semantic correctness, spelling, logical consistency, readability) on each subject.

Model	Strict	Relaxed
<i>Closed-source Models</i>		
Nano Banana Pro	70.8	92.8
GPT-Image-1.5	41.2	82.2
GPT-Image-1	12.6	62.9
Seedream 4.5	12.0	61.6
FLUX.2 max	10.4	62.4
Seedream 4.0	8.8	55.0
Imagen-4-Ultra	5.6	53.4
<i>Open-source T2I Models</i>		
FLUX.2 dev	0.4	42.1
Qwen-Image-2512	1.2	36.4
HiDream-I1-Full	0.0	21.2
Stable Diffusion 3.5 Large	0.0	16.0
<i>Open-source Unified MLLMs</i>		
BAGEL (thinking)	0.0	13.5
BAGEL	0.0	11.7
Show-o2-7B	0.0	12.0
Show-o2-1.5B-HQ	0.0	10.6
BLIP3o-NEXT-GRPO-Text-3B	0.0	12.6
BLIP3o-8B	0.0	6.8
Janus-Pro	0.0	9.8
Emu3	0.0	8.8

(a) GenExam-Mini

Model	Easy		Medium		Hard	
	Strict	Relaxed	Strict	Relaxed	Strict	Relaxed
<i>Closed-source Models</i>						
Nano Banana Pro	73.1	93.8	71.8	92.8	66.8	92.7
GPT-Image-1.5	50.8	85.6	41.4	79.1	38.3	81.3
GPT-Image-1	18.6	68.1	12.8	62.2	9.9	58.5
Seedream 4.5	12.8	59.9	9.8	56.9	14.5	61.9
FLUX.2 max	14.5	66.0	7.7	61.0	5.8	59.3
Imagen-4-Ultra	9.9	58.6	7.1	50.7	7.1	51.8
<i>Open-source T2I Models</i>						
FLUX.2 dev	2.9	48.9	2.1	40.4	2.4	39.9
Qwen-Image	0.4	34.6	0.3	25.4	0.0	21.3
HiDream-I1-Full	0.0	29.7	0.0	19.8	0.0	17.1
Stable Diffusion 3.5 Large	0.0	25.0	0.0	14.9	0.0	11.2
<i>Open-source Unified MLLMs</i>						
BAGEL (thinking)	0.0	21.1	0.0	12.8	0.0	7.7
BAGEL	0.0	17.9	0.0	11.5	0.0	7.1
Show-o2-7B	0.0	18.1	0.0	10.2	0.0	7.9
Show-o2-1.5B-HQ	0.0	17.1	0.0	8.7	0.0	6.9
BLIP3o-NEXT-GRPO-Text-3B	0.0	20.2	0.0	12.3	0.0	8.1
BLIP3o-8B	0.0	10.6	0.0	5.8	0.0	5.1
Janus-Pro	0.0	13.6	0.0	9.9	0.0	6.5
Emu3	0.0	14.2	0.0	7.5	0.0	6.0

(b) Subject Knowledge Difficulty

Table 6. Results on GenExam-Mini and on different subject knowledge difficulty levels.

Model	Chemical Structures		Diagrams		Geometric Shapes		Maps		Plots & Charts		Sheet Music		Trees & Graphs		Other	
	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel	Str	Rel
<i>Closed-source Models</i>																
Nano Banana Pro	51.2	85.2	71.6	94.5	56.5	84.2	85.7	97.6	82.6	96.3	61.7	90.8	63.9	89.3	65.5	89.8
GPT-Image-1.5	31.0	73.7	46.5	85.0	29.0	66.4	53.6	92.7	44.9	84.2	30.0	71.1	27.8	66.2	41.4	77.9
GPT-Image-1	7.1	45.5	15.6	66.3	7.3	54.3	8.0	68.5	14.1	64.3	10.0	54.0	8.3	52.0	13.8	61.1
Seedream 4.5	4.8	44.8	16.4	66.2	1.6	42.4	0.0	54.2	15.2	63.8	3.3	38.1	11.1	52.0	10.3	60.1
FLUX.2 max	6.0	46.1	9.7	67.2	8.1	56.1	7.1	67.6	6.5	57.5	6.7	48.5	11.1	51.4	10.3	60.8
Imagen-4-Ultra	6.0	35.6	10.5	60.4	3.2	37.9	5.4	57.7	7.2	52.5	0.0	39.5	0.0	32.2	6.9	47.9
<i>Open-source T2I Models</i>																
FLUX.2 dev	4.8	28.5	2.4	47.7	1.6	32.2	3.6	52.1	2.9	37.0	0.0	34.4	0.0	28.8	0.0	43.3
Qwen-Image	0.0	12.9	0.4	29.4	0.0	22.6	0.0	41.5	0.0	20.0	0.0	23.9	0.0	18.9	0.0	22.6
HiDream-I1-Full	0.0	10.3	0.0	22.7	0.0	21.2	0.0	34.2	0.0	17.7	0.0	21.9	0.0	17.0	0.0	20.0
Stable Diffusion 3.5 Large	0.0	8.1	0.0	17.7	0.0	13.1	0.0	29.3	0.0	9.1	0.0	24.9	0.0	7.8	0.0	11.2
<i>Open-source Unified MLLMs</i>																
BAGEL (thinking)	0.0	10.5	0.0	13.8	0.0	13.6	0.0	22.5	0.0	7.5	0.0	15.5	0.0	8.4	0.0	9.0
BAGEL	0.0	7.9	0.0	11.7	0.0	17.1	0.0	19.1	0.0	6.4	0.0	14.8	0.0	9.8	0.0	6.8
Show-o2-7B	0.0	4.3	0.0	13.1	0.0	10.6	0.0	22.8	0.0	7.2	0.0	8.9	0.0	3.8	0.0	9.3
Show-o2-1.5B-HQ	0.0	4.4	0.0	11.5	0.0	6.8	0.0	22.6	0.0	7.3	0.0	7.8	0.0	4.3	0.0	7.8
BLIP3o-NEXT-GRPO-Text-3B	0.0	7.8	0.0	12.9	0.0	16.6	0.0	16.7	0.0	8.6	0.0	16.4	0.0	13.4	0.0	14.9
BLIP3o-8B	0.0	4.3	0.0	7.2	0.0	8.1	0.0	16.5	0.0	2.6	0.0	6.5	0.0	3.4	0.0	5.6
Janus-Pro	0.0	9.3	0.0	9.5	0.0	15.4	0.0	11.9	0.0	4.3	0.0	15.0	0.0	4.7	0.0	11.4
Emu3	0.0	0.0	0.0	12.7	0.0	5.0	0.0	13.7	0.0	4.0	0.0	5.8	0.0	30.0	0.0	0.0

Table 7. Strict scores (Str) and relaxed scores (Rel) on GenExam for different image types.

GenExam: A Multidisciplinary Text-to-Image Exam

Level-2 Taxonomy	Nano Banana Pro	GPT-Image-1.5	GPT-Image-1	Seedream 4.5	FLUX.2 max	Imagen-4 Ultra	FLUX.2 dev	Qwen-Image	HiDream-I1-Full	BAGEL (thinking)	Show-o2-7B
Biology/Ecology	100.0	86.9	47.1	26.1	47.8	77.5	40.6	19.7	24.0	21.4	8.7
Biology/Genetics_and_Evolution	94.9	87.6	66.8	75.9	64.4	57.7	36.2	22.3	10.6	8.4	6.2
Biology/Physiology_and_Molecular_Process	97.8	92.5	73.0	75.0	75.4	62.1	47.8	23.2	20.5	9.3	8.6
Biology/Structure_and_Morphology	94.7	92.5	76.5	79.0	77.0	72.4	62.2	39.0	33.6	19.1	16.2
Chemistry/Chemical_Equilibrium	100.0	97.3	88.7	83.7	67.0	50.8	56.3	29.1	15.1	15.4	12.4
Chemistry/Chemical_Kinetics_and_Thermochemistry	99.0	97.5	87.8	76.8	86.4	73.5	62.3	34.9	22.1	11.6	1.4
Chemistry/Chemical_Reaction	83.4	66.3	34.2	34.1	34.5	30.2	20.0	3.5	4.9	2.3	0.3
Chemistry/Electrochemistry	100.0	87.8	74.4	71.6	75.6	70.4	39.9	24.1	16.8	12.3	2.2
Chemistry/Structure_Of_Matter	88.4	79.5	54.5	48.5	56.9	45.4	34.1	17.0	16.2	16.2	6.9
Computer/Data_Structures_and_Algorithms	88.4	67.4	52.1	51.2	49.0	34.0	27.4	20.3	17.4	7.6	3.9
Computer/Hardware_Architecture	96.0	77.4	48.2	62.1	63.3	41.7	36.6	20.1	12.2	2.4	3.0
Computer/Networking_and_Systems	95.9	83.1	59.8	61.6	57.3	50.6	35.0	10.9	12.9	1.9	2.7
Computer/Theory_and_AI	93.7	88.9	66.2	67.0	66.8	46.6	32.9	19.1	14.5	7.9	8.7
Economics/Finance_and_Decision	92.0	86.8	74.4	68.1	72.6	47.9	37.6	32.4	17.2	13.5	7.0
Economics/Macroeconomics	97.7	83.5	65.5	67.4	61.3	59.4	43.9	19.9	21.0	5.2	7.7
Economics/Microeconomics	97.3	88.2	64.8	67.1	60.0	62.1	44.2	18.9	15.9	6.7	6.1
Engineering/Civil_and_Architecture	95.8	87.5	66.4	67.5	67.3	64.7	51.3	34.9	30.2	13.9	16.0
Engineering/Mechanical_Engineering	93.8	87.5	64.4	69.8	69.7	68.3	55.5	38.0	27.4	15.5	11.5
Engineering/Mechanics_and_Dynamics	95.6	84.0	61.7	70.5	40.2	45.6	44.9	15.3	9.7	0.6	8.5
Engineering/Mechanics_of_Materials	96.1	89.5	69.8	68.4	68.4	49.9	33.8	17.2	21.8	8.7	4.7
Engineering/Surveying_and_Cartography	98.9	84.6	74.5	68.7	80.0	72.0	60.1	37.8	33.7	12.8	20.9
Engineering/Thermodynamics_and_Energy	94.5	84.7	63.1	71.6	74.4	72.9	43.4	32.4	20.0	6.6	10.3
Geography/Earth_Science	97.8	94.5	77.9	75.8	82.8	75.7	68.7	56.0	38.5	30.2	37.6
Geography/Human_and_Ecology	93.5	98.2	61.9	51.7	50.2	36.1	22.9	20.4	17.3	1.2	6.2
Geography/Maps	94.9	88.7	69.8	58.1	70.8	58.9	60.0	44.7	35.7	30.2	31.2
History/Historical_Data_Change	100.0	85.3	62.3	71.8	77.7	57.9	48.6	31.0	25.3	15.5	8.6
History/Historical_Map	99.8	94.8	68.6	50.1	64.5	56.4	47.4	40.5	33.9	16.3	15.9
History/Others	100.0	49.2	65.0	74.8	91.1	78.8	45.2	34.8	19.5	12.8	13.1
Mathematics/Analytic_Geometry	90.7	68.1	49.1	45.7	38.7	28.7	25.3	16.4	12.4	6.5	10.0
Mathematics/Plane_Geometry	83.7	63.0	52.1	43.4	54.0	36.2	33.8	18.2	18.3	14.4	11.0
Mathematics/Solid_Geometry	83.9	76.0	65.1	50.0	63.6	70.5	47.1	37.3	27.3	16.7	13.5
Music/Chord_Structures_and_Interval_Diagrams	93.4	52.4	36.6	48.1	36.4	21.2	19.1	8.1	13.5	1.0	1.0
Music/Key_Signatures_and_Time_Signatures	85.4	55.7	43.0	29.7	37.5	28.6	28.3	20.4	20.7	12.9	7.1
Music/Notes_and_Rests	91.5	77.2	57.7	37.3	50.9	43.7	37.3	27.2	23.1	17.9	10.7
Physics/Circuits_and_Electronics	90.2	73.2	55.1	55.6	47.1	39.1	29.1	14.7	8.8	7.2	4.6
Physics/Electromagnetism	97.5	91.9	71.9	66.1	77.4	66.0	53.0	35.1	26.0	24.6	18.2
Physics/Mechanics	99.4	92.9	74.9	68.2	76.7	71.5	50.7	37.7	23.6	14.0	16.2
Physics/Optics_and_Waves	95.2	87.7	74.5	74.9	71.1	69.8	54.7	28.8	24.8	20.7	19.0
Physics/Quantum_Mechanics	94.9	85.1	63.4	63.5	66.4	64.8	46.1	28.6	25.1	24.1	19.5
Physics/Thermodynamics	96.6	93.5	68.8	61.9	53.4	51.6	38.9	22.9	8.2	5.1	7.5

Table 8. Relaxed scores on level-2 taxonomy.

Subject	Dimension	Nano Banana Pro	GPT-Image-1.5	GPT-Image-1	Seedream 4.5	FLUX.2 max	Imagen-4-Ultra	FLUX.2 dev	Qwen-Image	HiDream-I1-Full	BAGEL (thinking)	Show-o2-7B
Biology	Semantic Correctness	0.96	0.91	0.72	0.74	0.78	0.69	0.59	0.32	0.26	0.1	0.12
	Spelling	1.9	1.95	1.53	1.72	1.05	1.21	0.7	0.28	0.25	0.17	0.04
	Logical Consistency	1.91	1.76	1.32	1.32	1.3	1.17	0.73	0.51	0.31	0.39	0.4
	Readability	1.98	1.94	1.74	1.74	1.69	1.63	1.28	1.15	1.24	1.01	0.37
Chemistry	Semantic Correctness	0.86	0.75	0.46	0.42	0.53	0.43	0.3	0.14	0.1	0.08	0.04
	Spelling	1.97	1.87	1.66	1.59	1.21	1.06	0.88	0.23	0.27	0.37	0.03
	Logical Consistency	1.72	1.37	0.86	0.69	0.72	0.57	0.46	0.14	0.12	0.37	0.21
	Readability	1.96	1.94	1.72	1.6	1.43	1.27	1.15	0.77	0.86	0.58	0.19
Economics	Semantic Correctness	0.97	0.84	0.64	0.65	0.64	0.61	0.44	0.2	0.18	0.05	0.07
	Spelling	2.0	1.94	1.66	1.88	1.35	1.34	1.06	0.39	0.19	0.0	0.01
	Logical Consistency	1.95	1.52	1.06	0.88	0.69	0.78	0.44	0.16	0.19	0.19	0.29
	Readability	1.97	1.82	1.53	1.57	1.29	1.27	1.1	0.69	0.88	0.36	0.14
Engineering	Semantic Correctness	0.95	0.84	0.62	0.68	0.71	0.67	0.5	0.33	0.19	0.08	0.11
	Spelling	1.95	1.97	1.64	1.79	1.29	1.36	1.0	0.5	0.66	0.23	0.12
	Logical Consistency	1.85	1.64	1.17	1.08	1.05	1.01	0.59	0.41	0.35	0.34	0.41
	Readability	1.95	1.86	1.64	1.56	1.46	1.41	1.19	0.92	1.15	0.48	0.31
Geography	Semantic Correctness	0.96	0.9	0.72	0.65	0.78	0.68	0.66	0.49	0.3	0.2	0.32
	Spelling	1.95	1.97	1.51	1.61	1.3	1.08	0.82	0.65	0.73	0.68	0.39
	Logical Consistency	1.89	1.88	1.41	1.12	1.32	1.18	1.03	0.91	0.82	0.91	0.92
	Readability	1.95	2.0	1.8	1.65	1.7	1.55	1.47	1.45	1.45	1.24	0.83
History	Semantic Correctness	1.0	0.89	0.66	0.51	0.71	0.6	0.52	0.41	0.32	0.11	0.14
	Spelling	2.0	2.0	1.41	1.56	1.1	0.83	0.56	0.39	0.27	0.15	0.12
	Logical Consistency	2.0	1.76	1.15	0.83	0.98	0.8	0.56	0.39	0.39	0.66	0.49
	Readability	2.0	1.95	1.63	1.49	1.66	1.46	1.12	1.24	1.2	0.9	0.34
Math	Semantic Correctness	0.84	0.6	0.43	0.37	0.46	0.3	0.26	0.13	0.09	0.04	0.08
	Spelling	1.97	1.95	1.77	1.73	1.53	1.38	1.28	0.81	0.83	0.53	0.18
	Logical Consistency	1.66	1.12	0.94	0.59	0.61	0.4	0.3	0.19	0.27	0.58	0.52
	Readability	1.89	1.7	1.62	1.45	1.3	1.16	1.08	0.9	0.93	0.59	0.38
Music	Semantic Correctness	0.9	0.64	0.42	0.28	0.36	0.27	0.23	0.13	0.08	0.07	0.05
	Spelling	1.98	1.97	1.74	1.62	1.63	1.51	1.38	0.91	0.88	0.86	0.57
	Logical Consistency	1.74	1.26	0.98	0.54	1.02	0.74	0.57	0.42	0.75	0.23	0.18
	Readability	1.92	1.89	1.89	1.57	1.71	1.63	1.54	1.49	1.54	0.86	0.28
Physics	Semantic Correctness	0.95	0.83	0.62	0.6	0.63	0.57	0.42	0.25	0.12	0.09	0.12
	Spelling	1.97	1.96	1.74	1.82	1.42	1.32	1.03	0.45	0.52	0.42	0.16
	Logical Consistency	1.84	1.65	1.16	0.85	0.88	0.83	0.5	0.42	0.25	0.42	0.31
	Readability	1.96	1.9	1.75	1.64	1.52	1.36	1.12	0.91	1.06	0.66	0.26
Computer Science	Semantic Correctness	0.91	0.72	0.48	0.53	0.54	0.38	0.27	0.16	0.11	0.03	0.05
	Spelling	1.98	1.94	1.75	1.84	1.49	1.25	1.1	0.66	0.52	0.29	0.05
	Logical Consistency	1.75	1.31	1.05	0.8	0.84	0.47	0.33	0.21	0.25	0.24	0.14
	Readability	1.93	1.84	1.66	1.51	1.42	1.04	0.97	0.75	0.81	0.29	0.11

Table 9. Scores of each dimension for all subjects.

D. Full Taxonomy List

In the table below, we show the complete list of four-level taxonomy and numbers of samples (denoted in parentheses). Blue text indicates mapping with ISCED-F (UNESCO, 2013) codes.

Level 1	Level 2	Level 3	Level 4
Biology (156)	Ecology (5) 0521 Environmental sciences	Ecosystem (5)	Food Chain and Food Web (4)
			Niche (1)
	Genetics and Evolution (13) 0511 Biology	Evolution and Population Genetics (3)	Evolutionary Tree (2)
			Pedigree Dominant and Recessive Genetic Diseases (1)
		Genetics (10)	Gene Linkage Map (1)
			Gene Structure (1)
			Mendelian Genetics (7)
			Transcription and Splicing Mechanisms (1)
		Cell Physiology (3)	Cell Division (3)
		Molecular Mechanism (4)	Metabolism (1)
			Microbiology (1)
		Molecular Mechanisms (15)	Signaling and Regulation (2)
			Genetic Information Transmission (12)
	Physiology and Molecular Process (49) 0511 Biology 0512 Biochemistry 0912 Medicine 0916 Pharmacy	Pathology and Pharmacology (6)	Material Metabolism (3)
			General Adaptation Syndrome (1)
			Infection Spread (2)
			Pharmacological Dose Response Curve (1)
		Systemic Physiology (21)	Stress Response Model (1)
			Tumor and Inflammation (1)
			Blood Coagulation (1)
			Germ Layer Differentiation (1)
			Immunity (3)
			Nervous System (13)
			Photosynthesis (2)
			Respiration and Gas Exchange (1)
		Cell Structure (27)	Basic Cell Structure (18)
			Microbial Morphology (4)
			Special Cells (5)
		Molecular Structure (6)	Biomacromolecules (4)
			Biomolecules (2)
	Structure and Morphology (89) 0511 Biology 0512 Biochemistry	Organ Structure (48)	Anal Canal (1)
			Bladder (1)
			Brain (16)
			Digestive Tract (3)
			Ear (1)
			Eye (4)
			Heart (4)
			Joint Structure (3)
			Kidney (2)
			Larynx (1)
			Lung (1)
			Muscular and Skeletal System (2)
			Nervous System (1)
			Pancreas (1)
			Plant Seed (1)
			Plant Stem Tissue (2)
			Skin (1)
			Vascular System (3)
		Others (2)	Optical Microscope Structure (2)
		Tissue Structure (6)	Epithelial Tissue (3)
			Plant Vascular Tissue (3)

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Level 1	Level 2	Level 3	Level 4
Chemistry (118) 0531 Chemistry	Chemical Equilibrium (5)	Acid Base Titration (2)	
		Solubility (3)	
	Chemical Kinetics and Thermochemistry (7)		
	Chemical Reaction (32)	Inorganic Reaction (2)	
		Organic Reaction (27)	
		Reaction Mechanism (3)	
	Electrochemistry (6)		
	Structure Of Matter (68)	Atomic Structure (14)	Atomic Model (6)
			Electron Configuration (8)
		Crystal Structure (3)	
		Element Abundance Distribution (2)	
		Molecular Structure (49)	Electron Configuration and Intermolecular Forces (8)
			Isomers (4)
			Molecular Structure Diagram (11)
			Organic Compound (26)
		ER Diagram (2)	
		Graph (24)	Adjacency List (3)
			Directed Graph (5)
			Graph Algorithms (4)
			Maximum Flow (1)
			Shortest Path (1)
			Undirected Graph (10)
Computer (102)	Data Structures and Algorithms (50) 0613 Software and applications development and analysis	Linked List (2)	
		Queue (1)	
		Sorting (2)	
		Tree (18)	B Tree (1)
			Others (2)
			Search Tree (7)
			Syntax Tree (2)
			Traversal (6)
		Bus Structure (3)	
		Cache (3)	Others (1)
	Hardware Architecture (17) 0613 Software and applications development and analysis	Digital Circuits (7)	Logic Gates (5)
			Multiplexer Application (2)
		Instruction Format (1)	
		Pipeline (3)	
	Networking and Systems (11) 0612 Database and network design and administration	Computer Networks (7)	Packet Structure (3)
			TCP and IP Protocol Stack (2)
			Topology (2)
		Operating System (4)	Deadlock Resource Allocation Diagram (1)
			Memory Paging (2)
			Process Scheduling (1)
		Compiler Principles (2)	
		Finite Automaton (8)	
	Theory and AI (24) 0613 Software and applications development and analysis 0619 Artificial Intelligence	Machine Learning (14)	Learning Rate Impact (2)
			Multicollinearity (1)
			Neural Networks (3)
			Overfitting and Underfitting Diagram (1)
			ROC Curve (1)
			Sampling Methods (1)
			Training and Testing Curves (4)
			Variance Bias Tradeoff (1)

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Level 1	Level 2	Level 3	Level 4
Economics (77)	Finance and Decision (5) 0412 Finance, banking and insurance	CAPM (2)	Beta Coefficient Scatter Plot (1)
		Risk Return Chart (3)	
		AD AS Model Aggregate Demand and Aggregate Supply (29)	
		Business Cycle (1)	
		Equilibrium Unemployment (1)	
	Macroeconomics (42) 0311 Economics	Exchange Rates and Monetary Policy (1)	
		IS LM Model (1)	
		Laffer Curve (1)	
		Lorenz Curve (1)	
		Other Economic Statistical Charts (3)	
		Other Macroeconomic Theories (1)	
		Phillips Curve Inflation and Unemployment (3)	
		Cost Curves (4)	
		Expected Utility Curve (1)	
		Game Theory (1)	
	Microeconomics (30) 0311 Economics	PPF and Utility Curves (7)	Budget Constraint Line (1)
			Indifference Curve Utility Maximization (6)
			Consumer and Producer Surplus (2)
			Elasticity Price and Income (4)
		Supply Demand Curves (17)	Equilibrium Price (9)
			Price Ceiling (1)
			Tax Impact (1)
Engineering (111)	Civil and Architecture (19) 0731 Architecture and town planning 0732 Building and civil engineering	Building Structure (4)	
			Detail Drawings (4)
		Engineering Drawings (11)	Plan View (1)
			Sectional and Profile Views (6)
		Geotechnical Engineering (3)	
	Mechanical Engineering (33) 0715 Mechanics and metal trades	Roads and Bridges (1)	
		Engineering Drawings (32)	Schematic Diagram (18)
			Section View (14)
	Mechanics and Dynamics (9) 0715 Mechanics and metal trades	Machine Parts (1)	Simple Parts (1)
		Vibration and Control (9)	Block Diagram (4)
			Damping and Vibration (5)
	Mechanics of Materials (11) 0715 Mechanics and metal trades		
		Stress Strain Relationship (11)	
	Surveying and Cartography (9) 0715 Mechanics and metal trades	Data Processing and Adjustment (3)	
		Maps and Diagrams (2)	
		Sectional and Profile Views (4)	
	Thermodynamics and Energy (30) 0713 Electricity and energy	Energy (2)	
		Energy Changes (5)	
		States and Phase Changes (10)	
		Thermal Cycle (13)	
Geography (66)	Earth Science (37) 0532 Earth sciences	Astronomy (3)	
			Atmospheric Circulation (2)
			Climate Chart (2)
		Climate and Circulation (16)	Energy Balance (1)
			Temperature Change (1)
			Water Cycle (10)
		Geological Age (1)	
		Geomagnetic Field (1)	
		Greenhouse Effect (1)	
		Landforms and Geology (14)	
		Latitude and Longitude (1)	

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Level 1	Level 2	Level 3	Level 4
History (41) 0222 History and archaeology	Human and Ecology (4) 0314 Sociology and cultural studies	Population and City (3)	
		Urban Planning Map (1)	
	Maps (25) 0314 Sociology and cultural studies	Earthquake Belt Distribution (2)	
		Tropical Temperate and Frigid Zones (2)	
		World and Regional Map (21)	
	Historical Data Change (7)	Others (2)	
		Population (3)	
		Unemployment Rate (2)	
	Historical Map (32)	Route Map (12)	Others (6)
		Territory Map (20)	Trade (6)
	Others (2)		
Mathematics (151) 0541 Mathematics	Analytic Geometry (56)	Absolute Value Function (1)	
		Definite Integral Area (12)	
		Exponential and Logarithmic Function (2)	
		Geometric Meaning Of Derivative (1)	
		Inequality Region (2)	Linear Programming (2)
		Linear Function (3)	
		Other Function (14)	
		Parametric Equation and Polar Curve (4)	
		Piecewise Function (8)	
		Quadratic Function (6)	
		Trigonometric Function (3)	
		Angle (3)	
		Circle (22)	Chord (8)
	Plane Geometry (84)		Inscribed and Circumscribed Circle (6)
			Others (4)
			Tangent (4)
		Complex Geometry Problem (23)	
			Other Polygon (2)
			Other Quadrilateral (5)
			Parallelogram (2)
			Pentagon (2)
			Rectangle (10)
			Regular Hexagon (1)
			Trapezoid (3)
			Altitude (1)
			Angle Bisector (1)
			Congruence (1)
			Others (1)
			Perpendicular Bisector (1)
			Right Triangle (4)
			Similarity (2)
	Solid Geometry (11)	Cylinder and Cone (4)	
			Oblique Prism (1)
			Right Prism (1)
			Regular Pyramid (2)
			Straight Cut (2)
			Tangent Plane (1)
Music (65) 0215 Music and performing arts	Chord Structures and Interval Diagrams (10)	Perfect Fifth and Major Third (3)	
	Key Signatures and Time Signatures (8)	Triads and Seventh Chords (7)	
		Circle Of Fifths (3)	
		Compound Triple Time (1)	
		Key Signature (4)	
	Notes and Rests (47)		

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Level 1	Level 2	Level 3	Level 4
Physics (113) 0533 Physics 0714 Electronics and automation	Circuits and Electronics (36)	Circuit Diagram (25)	Curves In Circuit (2)
			DC and AC Circuit (19)
			Kirchhoff Law Application (2)
			Op Amp Circuit (2)
		Component Symbols and Combinations (3)	Resistor (1)
			Transistor (2)
			Filter (1)
		Signal and Electronics (8)	Spectrum Diagram (2)
			Waveform Diagram (5)
	Electromagnetism (15)	Electric and Magnetic Field (8)	
		Electromagnetic Induction (7)	Ampere Force (1)
			Energy Loss and Hysteresis Loop (2)
			Lorentz Force (4)
	Mechanics (28)	Fluid Mechanics (23)	Gas Pressure (3)
			Liquid Pressure (15)
			Principle (5)
		Kinematics (4)	
		Newtonian Mechanics (1)	
	Optics and Waves (11)	Ray Diagram (9)	Interference and Diffraction (2)
			Lens Imaging (3)
			Reflection and Refraction (4)
		Wave (2)	
	Quantum Mechanics (10)	Atomic Structure (2)	
		Electromagnetic Wave (1)	
		Photoelectric Effect (2)	
		Potential Well (1)	
		Spectrum (4)	
	Thermodynamics (13)	Intermolecular Forces (2)	
		Molecular Speed Distribution (1)	
		Phase Change (8)	
		Pressure Volume Temperature Relationship (2)	

E. Prompts

E.1. Automatic Filtering

You are an expert-level visual reasoning evaluator. You will be given an image. Your task is to assess a given image for its suitability as a basis for multidisciplinary (i.e. requires subject knowledge such as math, physics, chemistry, biology, etc.) image generation. From each of the perspectives below, you should assign a score with a confidence score and a short explanation.

Perspectives

1. Text Richness Score (range: 0-10): If the image content is pure table/text without any other content (e.g. graphs, diagrams, plots, icons, geometry, etc.), it is 0. If the image does not contain any text, it is 10. For other cases, the score is based on the complexity of the image content.
2. Image Domain (range: 0-1): If the image is a natural image (i.e., real image) / pathological image (also including body scans, MRI, CT scans, and X-rays), it is 0, otherwise it is 1.
3. Image Complexity (range: 1-10): Whether the image is too complex to draw (consider the number and complexity of components, objects, text, etc.), very complex is 1, very simple is 10.
4. Subject Knowledge (range: 1-10): Whether the model needs **subject knowledge and reasoning** to generate this image (note: the prompt is not a complete description of the image, it is related to subject knowledge), no need is 1, need a lot is 10.

Instructions

For each domain, provide:

- **Score**: An integer, based on the range defined above.
- **Confidence Score**: An integer between 1 and 5.
- **Short Explanation**: 1-2 sentences justifying the score, referencing the image content and its alignment with the domain definition. Make sure your explanation is concise.

Output Format

Respond **STRICTLY** in the following format:

```
{
  "Text Richness Score": {
    "score": 0,
    "confidence": 0,
    "explanation": ""
  },
  "Image Domain": {
    "score": 0,
    "confidence": 0,
    "explanation": ""
  },
  "Image Complexity": {
    "score": 0,
    "confidence": 0,
    "explanation": ""
  },
  "Subject Knowledge": {
    "score": 0,
    "confidence": 0,
    "explanation": ""
  }
}
```

Make sure your response follows the format strictly and can be parsed by Python `json.loads`. Do not include any other explanation.

E.2. Annotating

You are an expert-level prompt generator for multidisciplinary text-to-image generation. You are given an image, a related conversation between a user and a model, and its corresponding taxonomy.

Your job is to provide a **prompt** (like an academic question) for multidisciplinary text-to-image generation corresponding to this image. You will also generate some **scoring points** when evaluating whether some model generates the correct image corresponding to this prompt.

Instructions

Prompt

1. The prompt should **NOT** be a complete description of the image. It should be in a style similar to an **academic question** that **requires** some subject knowledge and reasoning to generate the image. If a model does not have such subject knowledge, it is expected to fail to generate the correct image.
2. The prompt should be precise and concise (less than 200 words) and should be in English.
3. The provided conversation and text in the image can sometimes help you to generate the prompt, but you should consider them carefully since they may contain some irrelevant information.

Scoring Points

1. The scoring points are in the form of a list of questions and their corresponding scores, answered by “Yes” or “No” only. They are designed so that by answering the questions, we can evaluate whether the model can generate the correct image corresponding to the prompt. It is expected that the answers are all “Yes” if the model generates the correct image.
2. The number of scoring points should be no more than 20. If the image is simple and requires little subject knowledge and reasoning, the number of scoring points can be small.
3. The questions should be based on the prompt instead of the image, i.e. since the model is required to generate the image based on the prompt, the questions should not focus on information or components that are in the ground truth image but not in the prompt.
4. The questions should not be too general. It should provide **details** rather than “Is xxx correct?”, e.g. the structure of a molecule (bonds, position of atoms, etc.), characteristics that a curve should satisfy, etc. It should not be too complex as you should break it down into several sub-questions.
5. The score of each scoring point should be a number between 0 and 1, indicating the proportion of this question in the total score of the image. **The sum of all scores should be 1**.
6. Typically, the score of the most basic question (e.g. checking the overall structure of the image) should be small, such as 0.1 or 0.2. For complex questions, questions requiring subject knowledge, or questions about details, the score should be large.

Examples

1. Prompt: “Generate a planar molecular structure diagram of benzene (C_6H_6), showing its conjugated double bond structure, and distinguish carbon atoms and hydrogen atoms with different colors.”, Scoring points: $\{ \{ \text{“question”: “Is the number of carbon atoms 6?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Is the number of hydrogen atoms 6?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Is the number of double bonds 3?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Is the number of single bonds 3?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Are carbon atoms and hydrogen atoms in different colors?”}, \text{“score”: 0.2} \} \}$
2. Prompt: “Generate the graph of the function $y = e^x$.”, Scoring points: $\{ \{ \text{“question”: “Does the image shows a graph of a function?”}, \text{“score”: 0.1} \}, \{ \text{“question”: “Does the curve pass through the point (0, 1)?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Does the curve asymptotically approach the x-axis when x approaches negative infinity?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Does the curve tends to positive infinity when x approaches positive infinity?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Is the curve smooth and continuous?”}, \text{“score”: 0.1} \}, \{ \text{“question”: “Is the slope always positive and increasing when x increases?”}, \text{“score”: 0.2} \} \}$
3. Prompt: “Generate a schematic diagram of an animal cell structure. Label cell membrane, cytoplasm, nucleus, and mitochondria.”, Scoring points: $\{ \{ \text{“question”: “Does the image contain a cell structure?”}, \text{“score”: 0.1} \}, \{ \text{“question”: “Does the cell membrane label correspond to the correct position in the cell?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Does the cytoplasm label correspond to the correct position in the cell?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Does the nucleus label correspond to the correct position in the cell?”}, \text{“score”: 0.2} \}, \{ \text{“question”: “Does the mitochondria label correspond to the correct position in the cell?”}, \text{“score”: 0.3} \} \}$

Output Format

Respond **STRICTLY** in the following format:

```
{ {
  "prompt": "",
  "scoring_points": [
    { "question": "question1", "score": 0.1 },
    ...
  ],
}
```

Make sure your response follows the format strictly and can be parsed by Python `json.loads`. Do not include any other explanation.

Taxonomy of this image

```
{ taxonomy }
```

Conversation

```
{ conversations }
```

E.3. Evaluation

You are an expert-level evaluator for multidisciplinary text-to-image generation. You will be given an input prompt for text-to-image generation, an image generated by a model (the first image), a ground truth image for reference (the second image), and several scoring point questions and their corresponding scores. Your task is to evaluate the generated image by answering the scoring point questions and evaluating the image from some global perspectives.

Instructions

1. Remember that the first image is the generated image from a model, and the second image is the ground truth image. You should **only** evaluate the generated image, while the ground truth image can be used for reference.
2. First, give a detailed description of all the components in the generated image.
3. For each scoring point, you should use the ground truth image as **reference information** to make more accurate judgments. You should generate a **detailed** reasoning **step by step**. It should first analyze the question, what subject knowledge it requires and what information you need to refer to from the ground truth image, then analyze whether the generated image satisfies the scoring point based on the scoring point, prompt and reference information. Finally, answer 1 if the generated image fully satisfies the scoring point, otherwise answer 0. For example, if a scoring point is "Does the cell membrane label correspond to the correct position in the cell?", you can refer to the ground truth image to check the correct position of the cell membrane label as an example, by **explicitly** analyzing the ground truth image in your reasoning.
4. You should also evaluate the generated image from some global perspectives provided below. For each perspective, first provide a **detailed step-by-step** reasoning, e.g. recognize all the text labels for spelling, then give a score in the defined range.

Global Perspectives

1. Spelling (range: 0-2): The spelling of the text in the image, including the notations and equations. You should first recognize the text in the image in the reasoning, then check the spelling of the text. Specifically:

- 0: There are critical errors in spelling, notations or equations which significantly hinders the understanding of the image.
- 1: There are some errors in spelling, notations or equations that somehow hinder the understanding of key information in the image.
- 2: All or almost all the spelling, notations and equations are correct. Tiny errors like commas, capital letters, etc. are allowed.

2. Readability (range: 0-2): The readability of the image. Each component in the image should be clearly readable and identifiable. All text labels and marks should in the right place and are not overlapped or occluded by other elements. If the image is geometry or diagram, there should not be unlabeled points or lines or duplicated labels. If the image is a plot or chart, the axis should be labeled and the ticks should be carefully checked. You should first identify the components, labels, and marks in the image in the reasoning, then check their readability. Specifically:

- 0: There are critical readability issues which significantly hinders the understanding of the image e.g. some components are impossible to distinguish, or some labels are fully overlapped or occluded, or many key information are not labeled.
- 1: There are some readability issues, e.g. some components are not clearly readable and identifiable, or some labels are overlapped or occluded, or some key information are not labeled.
- 2: The readability is perfect or almost perfect. Components are clearly readable and identifiable and necessary labels and marks are present. Tiny errors are allowed.

3. Logical Consistency (range: 0-2): The logical consistency of the image. Check the correctness of all the marks, text, musical notes, etc. If the image is geometry, check the correctness of each marked angles, lengths, coordinates, etc. If the image is a plot or chart, check the correctness of each data point, the axis, the ticks, the legend, etc. You should first identify the components, labels, and marks in the image in the reasoning, then check their logical consistency. Specifically:

- 0: There are critical logical consistency issues which significantly hinders the understanding of the image, e.g. some key marks are not correct, some angles/lengths/data points are significantly inconsistent with the text label, etc.
- 1: There are some logical consistency issues that somehow hinder the understanding of key information in the image, e.g. some marks are not correct, some angles/lengths/data points are inconsistent with the text label, etc.
- 2: The logical consistency is perfect or almost perfect. Marks, text, etc. are correct and consistent. Tiny errors are allowed.

Output Format

Respond **STRICTLY** in the following format:

```
{
  "description": "...",
  "answers": [
    {
      "reasoning": "...",
      "answer": 1
    },
    ...
  ],
  "global_evaluation": {
    "Spelling": {
      "reasoning": "...",
      "score": 0,
    },
    "Readability": {
      "reasoning": "...",
```

```

    "score": 0,
  },
  "Logical Consistency": {
    "reasoning": "...",
    "score": 0,
  },
}
}

```

Make sure your response follows the format strictly and can be parsed by Python json.loads. Do not include any other explanation.

Prompt

{prompt}

Scoring Points

{scoring_points}

F. More Visualization

More visualization of generated images are given in Figs. 11, 12 and 13.

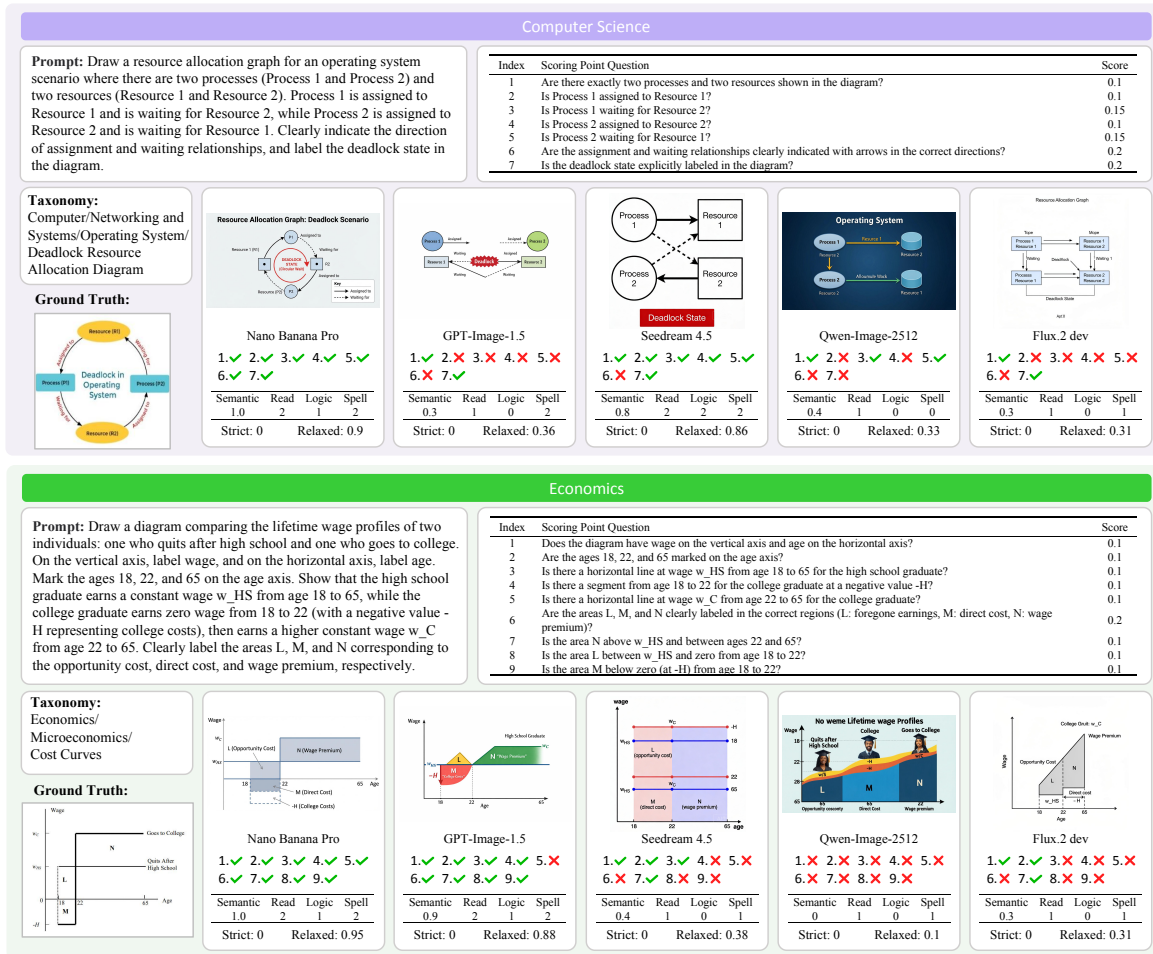


Figure 11. Visualization of generated images.

GenExam: A Multidisciplinary Text-to-Image Exam



Geography

Prompt: Draw a cross-sectional diagram illustrating the process of cavern formation in a limestone region, showing the unsaturated zone, water table, groundwater zone, and the direction of groundwater flow. Indicate where caverns are forming and the presence of a surface stream above the cavern. Label all key features.

Taxonomy: Geography/Earth Science/
Landforms and Geology

Ground Truth:

Index	Scoring Point Question	Score
1	Is there a distinct unsaturated zone above a saturated groundwater zone separated by a water table boundary?	0.3
2	Is there a surface stream depicted above the cavern location, as required by the prompt?	0.15
3	Is the direction of groundwater flow clearly indicated in the saturated (groundwater) zone with flowlines or arrows?	0.35
4	Are caverns shown forming within the limestone, and is their location indicated with a surface stream positioned above them?	0.15
5	Is the overall scene a lateral cross-section through karst bedrock rather than a plan view or non-karst landscape?	0.05

Nano Banana Pro

1.✓ 2.✓ 3.✓ 4.✓ 5.✓

Semantic	Read	Logic	Spell
1.0	2	2	2

Strict: 1 Relaxed: 1.0

GPT-Image-1.5

1.✓ 2.✓ 3.✓ 4.✓ 5.✓

Semantic	Read	Logic	Spell
1.0	2	2	2

Strict: 1 Relaxed: 1.0

Seedream 4.5

1.✓ 2.✓ 3.✓ 4.✓ 5.✓

Semantic	Read	Logic	Spell
1.0	2	2	1

Strict: 0 Relaxed: 0.95

Qwen-Image-2512

1.✓ 2.✓ 3.✗ 4.✓ 5.✓

Semantic	Read	Logic	Spell
0.65	2	1	1

Strict: 0 Relaxed: 0.655

Flux.2 dev

1.✓ 2.✓ 3.✓ 4.✓ 5.✓

Semantic	Read	Logic	Spell
1.0	1	1	1

Strict: 0 Relaxed: 0.85

Music

Prompt: Create an instructional poster for music theory that demonstrates transposition using the Circle of Semitones. Draw the Circle of Semitones labeled with major-key tonics around the circle in correct order. Also annotate two specific interval moves on the circle: from D to E is up a whole step (clockwise), and from B to G is down a major third (counterclockwise). Beneath the circle, show two staff examples: (1) treble clef D major (two sharps) transposed up a whole step to treble clef E major (four sharps); (2) treble clef B minor (two sharps) transposed down a major third to bass clef G minor (two flats), with arrows between each pair to show the direction of transposition.

Taxonomy: Music/Key Signatures/
Time Signatures/
Circle Of Fifths

Ground Truth:

Index	Scoring Point Question	Score
1	Are the key names arranged on a Circle of Semitones in the correct clockwise order?	0.33
2	Is the interval from D to E explicitly depicted as an upward whole step on the circle (clockwise)?	0.1
3	Is the interval from B to G depicted as a downward major third on the circle (counterclockwise)?	0.28
4	Does the first staff example show treble clef D major with two sharps?	0.1
5	Does the transposed partner show treble clef E major with four sharps?	0.1
6	In the second example, is B minor shown on treble clef with two sharps?	0.05
7	Is the transposed partner G minor shown on bass clef with two flats?	0.05

Nano Banana Pro

1.✗ 2.✓ 3.✓ 4.✓ 5.✓
6.✓ 7.✓

Semantic	Read	Logic	Spell
0.675	2	1	2

Strict: 0 Relaxed: 0.723

GPT-Image-1.5

1.✗ 2.✓ 3.✓ 4.✓ 5.✓
6.✗ 7.✓

Semantic	Read	Logic	Spell
0.625	2	1	2

Strict: 0 Relaxed: 0.688

Seedream 4.5

1.✗ 2.✗ 3.✗ 4.✗ 5.✗
6.✗ 7.✓

Semantic	Read	Logic	Spell
0	2	0	1

Strict: 0 Relaxed: 0.15

Qwen-Image-2512

1.✗ 2.✗ 3.✗ 4.✓ 5.✓
6.✓ 7.✓

Semantic	Read	Logic	Spell
0.3	0	0	0

Strict: 0 Relaxed: 0.21

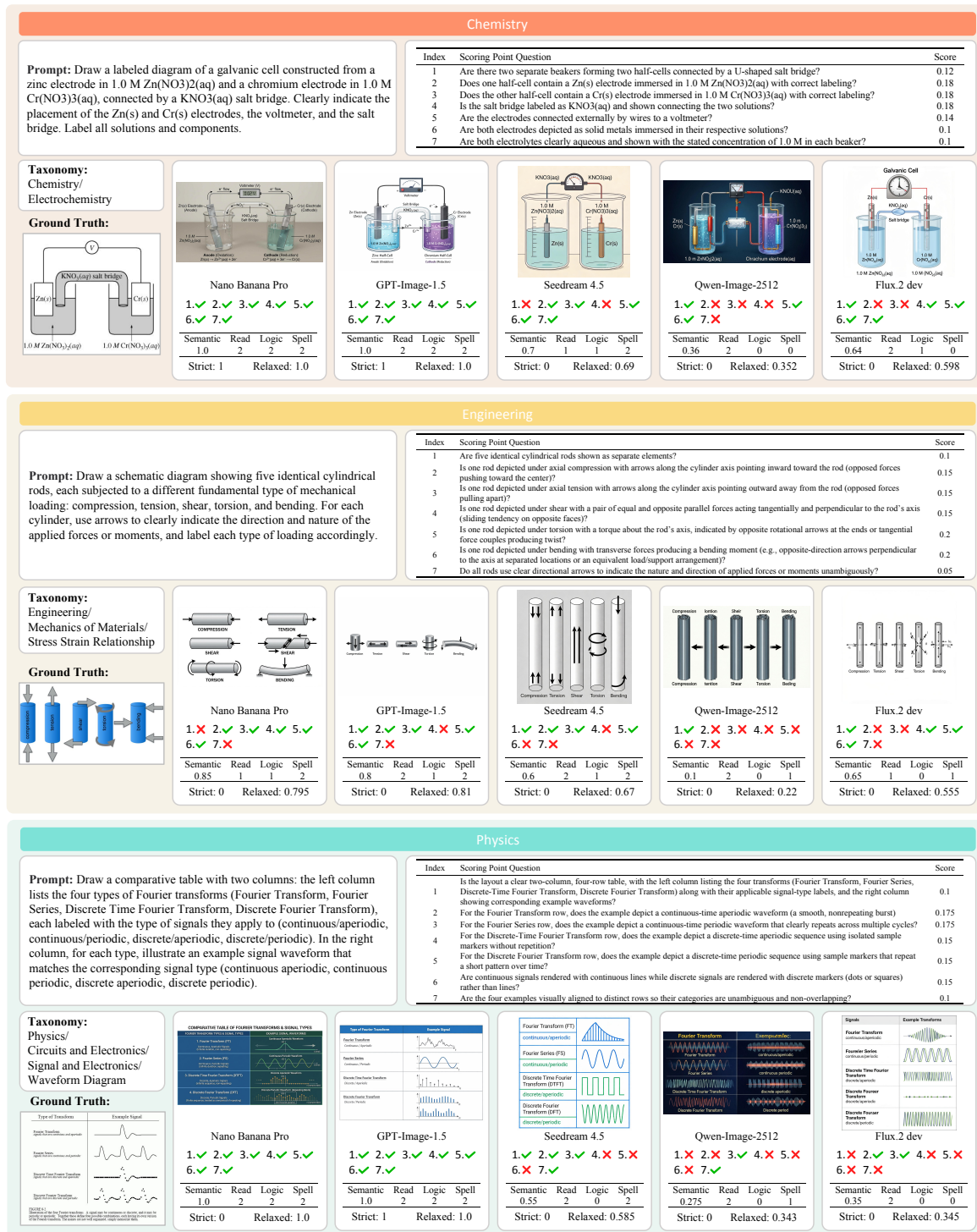
Flux.2 dev

1.✗ 2.✗ 3.✗ 4.✓ 5.✗
6.✗ 7.✓

Semantic	Read	Logic	Spell
0.15	1	0	0

Strict: 0 Relaxed: 0.155

Figure 12. Visualization of generated images.



Physics

Prompt: Draw a comparative table with two columns: the left column lists the four types of Fourier transforms (Fourier Transform, Fourier Series, Discrete Time Fourier Transform, Discrete Fourier Transform), each labeled with the type of signals they apply to (continuous/aperiodic, continuous/periodic, discrete/aperiodic, discrete/periodic). In the right column, for each type, illustrate an example signal waveform that matches the corresponding signal type (continuous aperiodic, continuous periodic, discrete aperiodic, discrete periodic).

Index	Scoring Point Question	Score
1	Is the layout a clear two-column, four-row table, with the left column listing the four transforms (Fourier Transform, Fourier Series, Discrete-Time Fourier Transform, Discrete Fourier Transform) along with their applicable signal-type labels, and the right column showing corresponding example waveforms?	0.1
2	For the Fourier Transform row, does the example depict a continuous-time aperiodic waveform (a smooth, nonrepeating burst)?	0.175
3	For the Fourier Series row, does the example depict a continuous-time periodic waveform that clearly repeats across multiple cycles?	0.175
4	For the Discrete-Time Fourier Transform row, does the example depict a discrete-time aperiodic sequence using isolated sample markers without repetition?	0.15
5	For the Discrete Fourier Transform row, does the example depict a discrete-time periodic sequence using sample markers that repeat a short pattern over time?	0.15
6	Are continuous signals rendered with continuous lines while discrete signals are rendered with discrete markers (dots or squares) rather than lines?	0.15
7	Are the four examples visually aligned to distinct rows so their categories are unambiguous and non-overlapping?	0.1

Taxonomy:
Physics/
Circuits and Electronics/
Signal and Electronics/
Waveform Diagram

Ground Truth:

Nano Banana Pro

1. ✓ 2. ✓ 3. ✓ 4. ✓ 5. ✓
6. ✓ 7. ✓

Semantic: 1.0 Read: 2 Logic: 2 Spell: 2

Strict: 0 Relaxed: 1.0

GPT-Image-1.5

1. ✓ 2. ✓ 3. ✓ 4. ✓ 5. ✓
6. ✓ 7. ✓

Semantic: 1.0 Read: 2 Logic: 2 Spell: 2

Strict: 1 Relaxed: 1.0

Seedream 4.5

1. ✓ 2. ✓ 3. ✓ 4. ✗ 5. ✗
6. ✗ 7. ✓

Semantic: 0.55 Read: 2 Logic: 0 Spell: 2

Strict: 0 Relaxed: 0.585

Qwen-Image-2512

1. ✗ 2. ✗ 3. ✓ 4. ✗ 5. ✗
6. ✗ 7. ✗

Semantic: 0.275 Read: 2 Logic: 0 Spell: 1

Strict: 0 Relaxed: 0.343

Flux 2 dev

1. ✗ 2. ✓ 3. ✓ 4. ✗ 5. ✗
6. ✗ 7. ✗

Semantic: 0.35 Read: 2 Logic: 0 Spell: 0

Strict: 0 Relaxed: 0.345

Figure 13. Visualization of generated images.